

Lecture

Internet Trends and Web Basics

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What are We Talking About Today?

- **Internet vs. World Wide Web**

- Internet: Global digital infrastructure
- WWW: System for accessing web content
- Intranet/Extranet: Internal and external networks

- **Recent Internet Trends**

- Growth in users, smartphones, and digital data
- Rise in social media and cloud usage
- Mobile surpassing desktop for internet access

- **Web Technologies & History**

- Client/Server Architecture: Browsers and servers
- Core Technologies: HTTP, URLs, HTML
- Key Milestones: Development of WWW, major browsers, companies like Google and YouTube.

The Internet and the WWW are Different

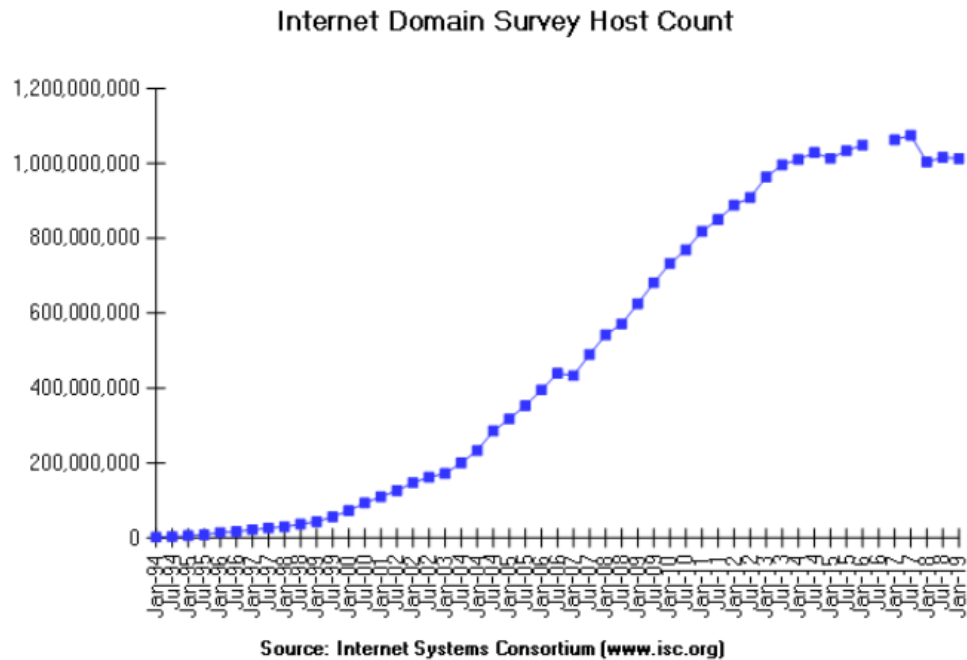
- The *Internet* is a global digital infrastructure that connects hundreds of millions of computers and people
- The *World Wide Web* is a mechanism that unifies the retrieval and display of a subset of data on the Internet
- An *intranet* is a local/global information structure that connects an organization internally. Intranets today often make use of Web technologies
- An *extranet* is a private network that uses the public telecommunication system to securely share part of a business's information or operations with suppliers, vendors, partners, customers, or other businesses.

Recent Trends in Internet Development

- Growth in number of users connected
- Growth in Smartphone use, particularly iOS and Android
- Growth in digital data, especially photos and video
- Growth in Social Media
- Growth in Internet use from Mobile over desktop/laptop
- Growth in tablet usage over desktops/laptops
- Decreased dominance of Microsoft Windows
- Growth in use of the cloud

How Big is the Internet (historical)

<https://www.isc.org/network/survey>

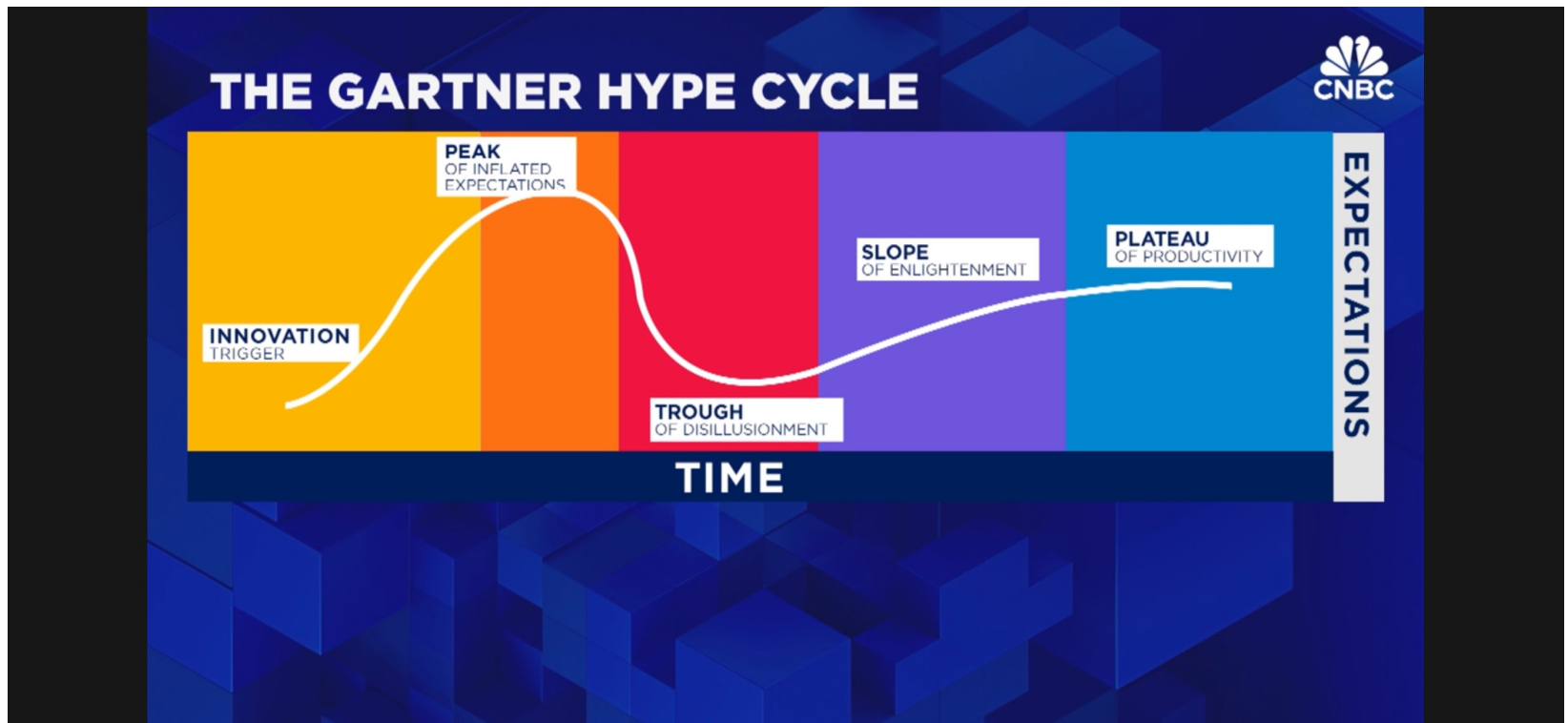


hosts were doubling every 18 months, but growth has slowed
See the survey background at: <http://www.isc.org/network/survey>

It counts the number of IP addresses that have been assigned a name. The survey queries the domain name system for the name assigned to every possible IP address. But rather than sending a query to every one of the 4.3 billion possible IP addresses, the survey starts with a list of all network numbers that have been delegated within the IN-ADDR.ARPA domain. **This survey has been discontinued.**

Date	HostCount
Jan 19	1,012,695,272
Jul 18	1,015,787,389
Jan 18	1,003,604,363
Jul 17	1,074,971,748
Jan 17	1,062,660,523
Jan 16	1,048,766,623
Jul 15	1,033,836,245
Jan 15	1,012,706,608
Jul 14	1,028,544,414
Jan 14	1,010,251,829
Jul 13	996,230,757
Jan 13	963,518,598
Jul 12	908,585,739
Jan 12	888,239,420
Jul 11	849,869,781
Jan 11	818,374,269
Jul 10	768,913,036
Jan 10	732,740,444
Jul 09	681,064,561
Jan 09	625,226,456
Jul 08	570,937,778
Jan 08	541,677,360
Jul 07	489,774,269
Jan 07	433,193,199
Jul 06	439,286,364
Jan 06	394,991,609
Jul 05	353,284,187
Jan 05	317,646,084
Jul 04	285,139,107
Jan 04	233,101,481
Jan 03	171,638,297
Jul 02	162,128,493
Jan 02	147,344,723
Jul 01	125,888,197
Jan 01	109,574,429
Jul 00	93,047,785
Jan 00	72,398,092
Jul 99	56,218,000
Jan 99	43,230,000
Jul 98	36,739,000
Jan 98	29,670,000
Jul 97	19,540,000
Jan 97	16,146,000
Jul 96	12,881,000
Jan 96	9,472,000
Jul 95	6,642,000
Jan 95	4,852,000
Jul 94	3,212,000
Jan 94	2,217,000
Jul 93	1,776,000
Jan 93	1,313,000

Gartner Hype Cycles



Source: <https://www.cnbc.com/video/2024/07/19/ais-trillion-dollar-time-bomb.html>

Gartner Hype Cycles (cont'd)

- **Innovation Trigger**

- A potential technology breakthrough kicks things off. Early proof-of-concept stories and media interest trigger significant publicity. Often no usable products exist and commercial viability is unproven.

- **Peak of Inflated Expectations**

- Early publicity produces a number of success stories — often accompanied by scores of failures. Some companies take action; many do not.

- **Trough of Disillusionment**

- Interest wanes as experiments and implementations fail to deliver. Producers of the technology shake out or fail. Investments continue only if the surviving providers improve their products to the satisfaction of early adopters.

Gartner Hype Cycles (cont'd)

- **Slope of Enlightenment**

- More instances of how the technology can benefit the enterprise start to crystallize and become more widely understood. Second- and third-generation products appear from technology providers. More enterprises fund pilots; conservative companies remain cautious.

- **Plateau of Productivity**

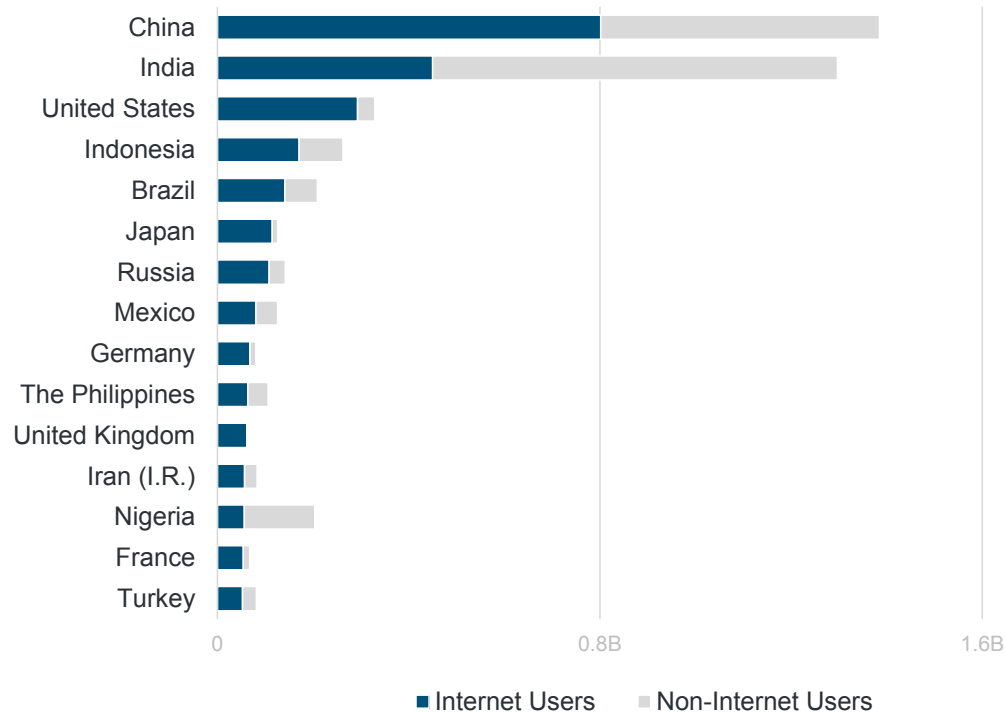
- Mainstream adoption starts to take off. Criteria for assessing provider viability are more clearly defined. The technology's broad market applicability and relevance are clearly paying off.

Source:

<https://www.gartner.com/en/research/methodologies/gartner-hype-cycle>

Global Internet Users =
China @ 21% of Total...India @ 12%...USA @ 8%

Internet Users – Top Countries, 2018



The following slides are based upon a presentation by Mary Meeker of Bond and formerly of Kleiner Perkins Caufield and Byers, see <https://www.bondcap.com/#internettrends>

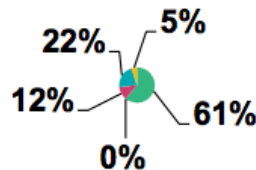
Internet Users – 1995 → 2014...

<1% to 39% Population Penetration Globally

1995

35MM+ Internet Users

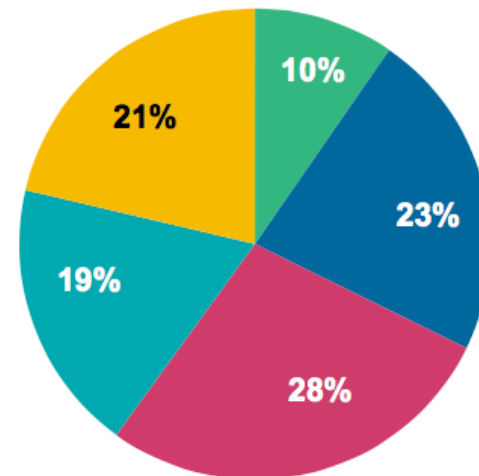
0.6% Population Penetration



2014

2.8B Internet Users

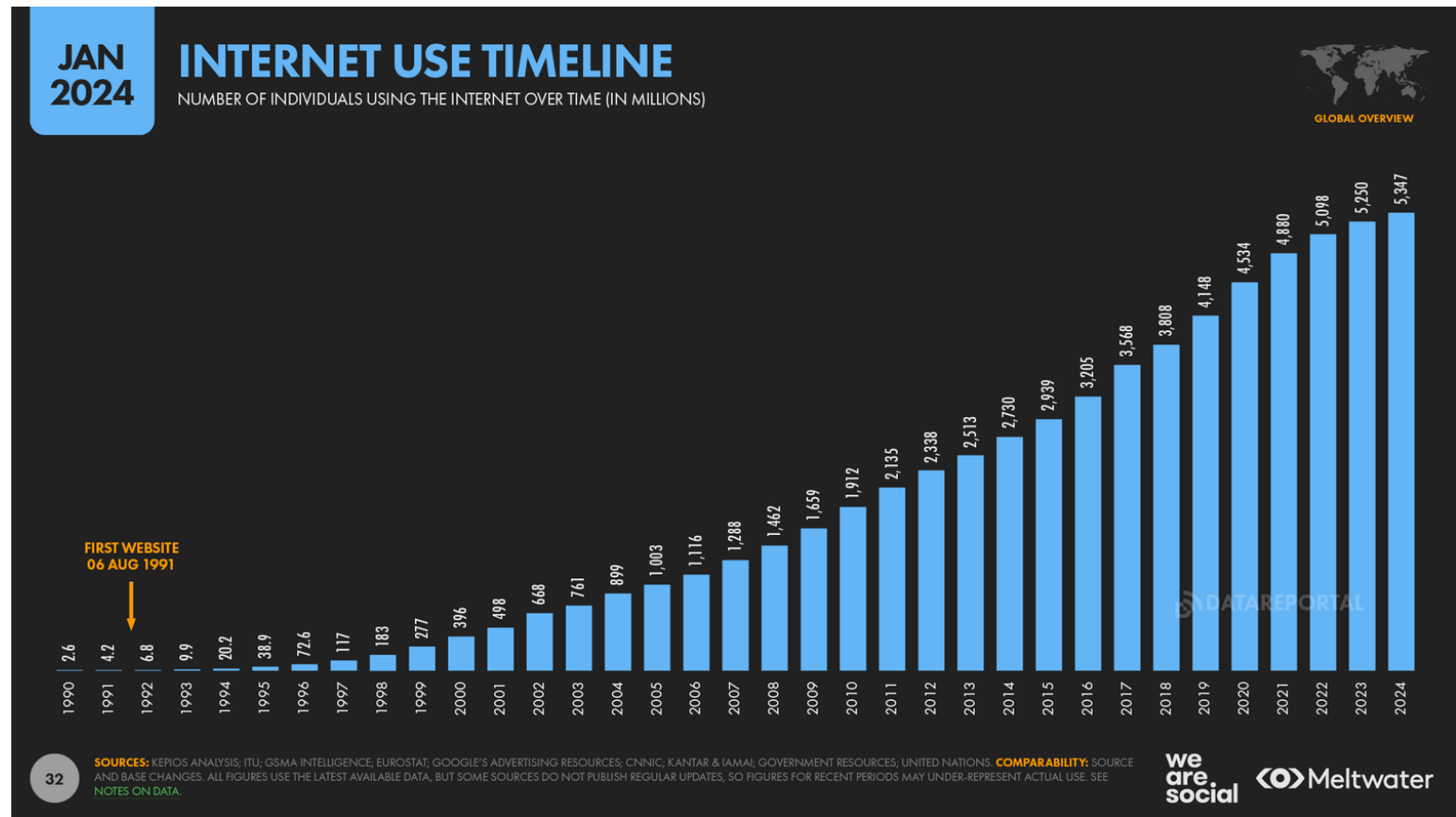
39% Population Penetration



■ USA ■ China ■ Asia (ex. China) ■ Europe ■ Rest of World

Internet Users 2024

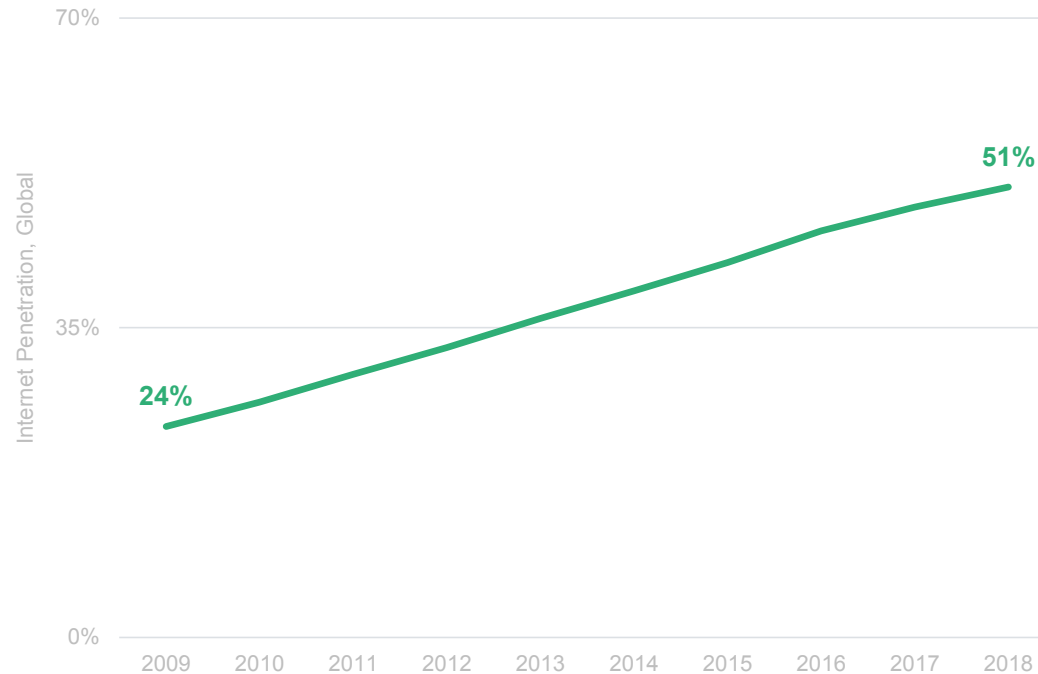
5.35 billions or 66.2% of global population



Source: <https://datareportal.com/reports/digital-2024-deep-dive-the-state-of-internet-adoption>

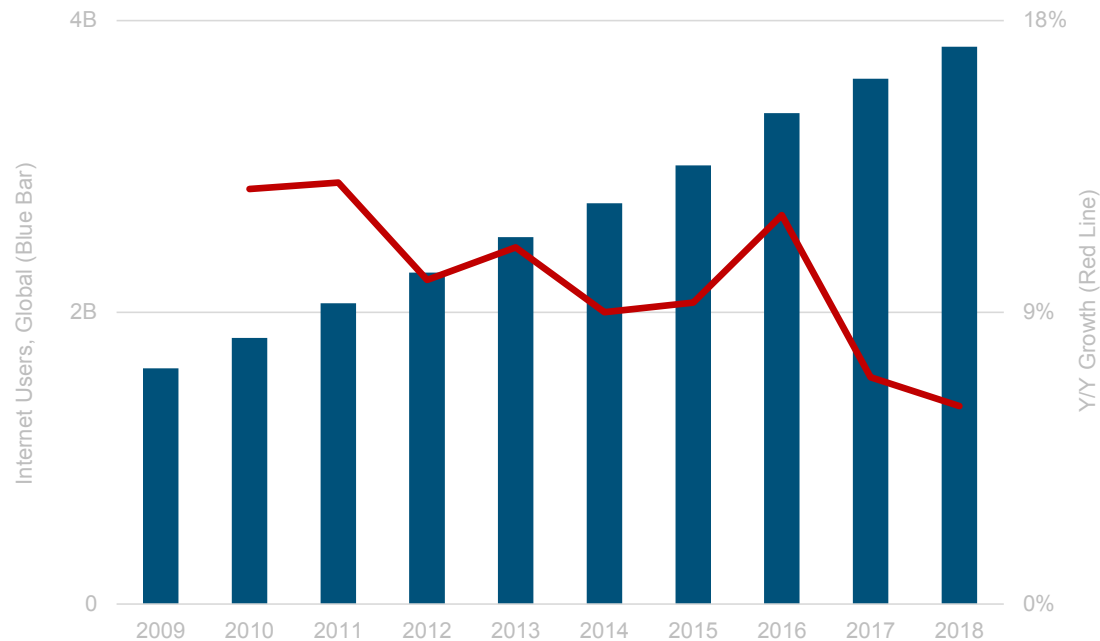
Global Internet Users =
3.8B >50% of Population

Internet Penetration, 2018



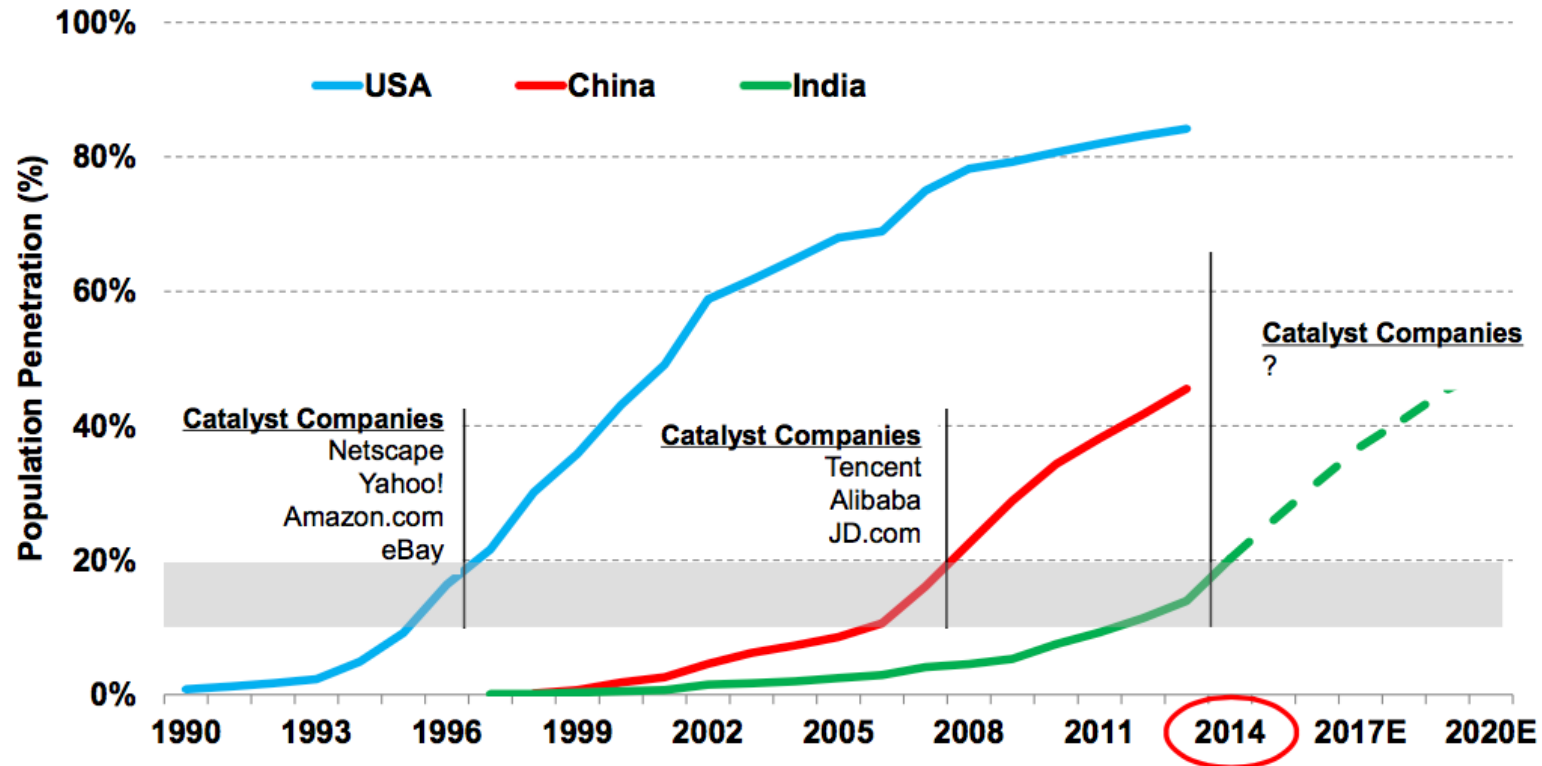
Global Internet User Growth = Solid But Slowing +6% vs. +7% Y/Y

Internet Users vs. Y/Y Growth



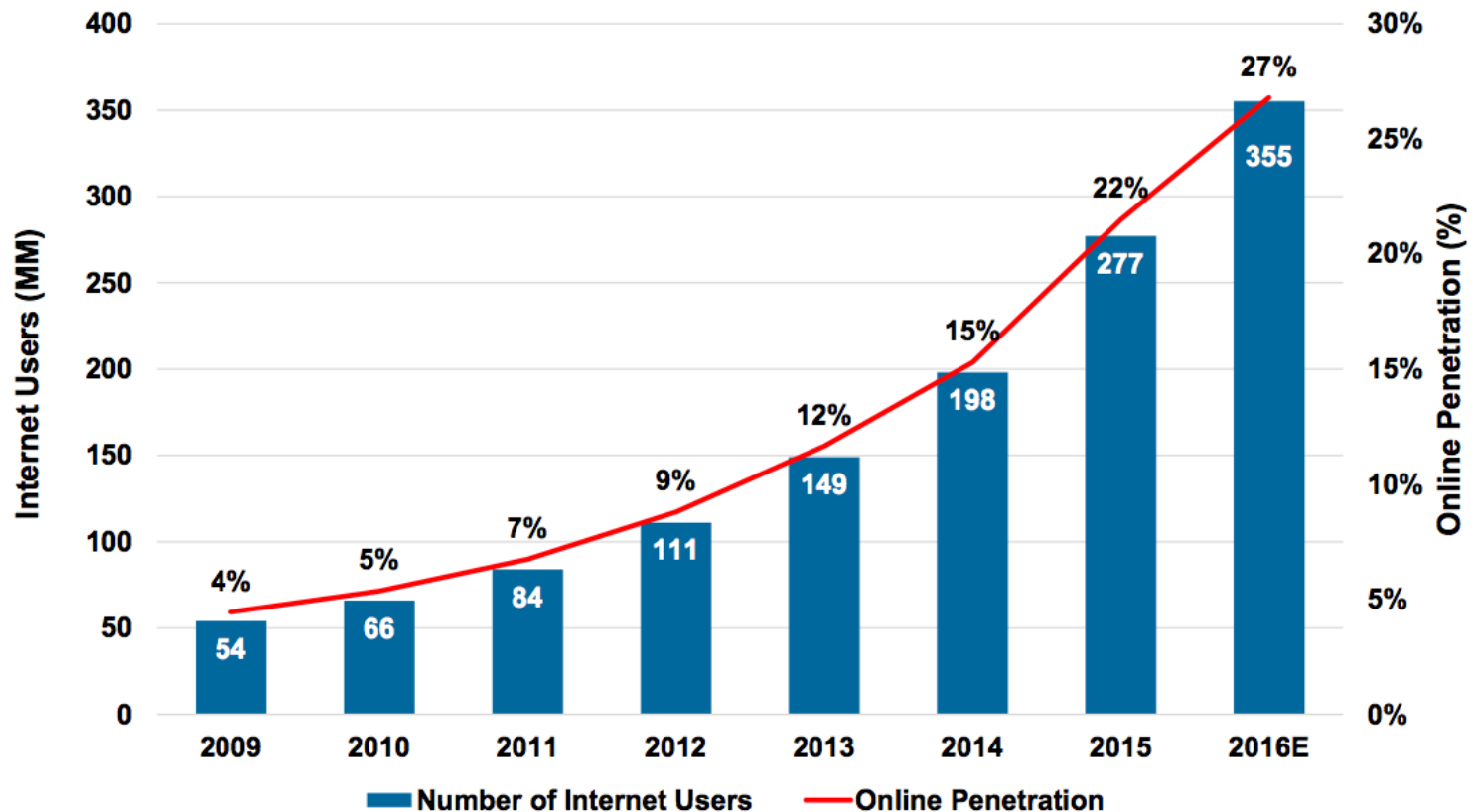
India = Appears to Be @ Internet Penetration Growth Inflection

Internet User Penetration Curve, USA / China / India, 1990 – 2020E



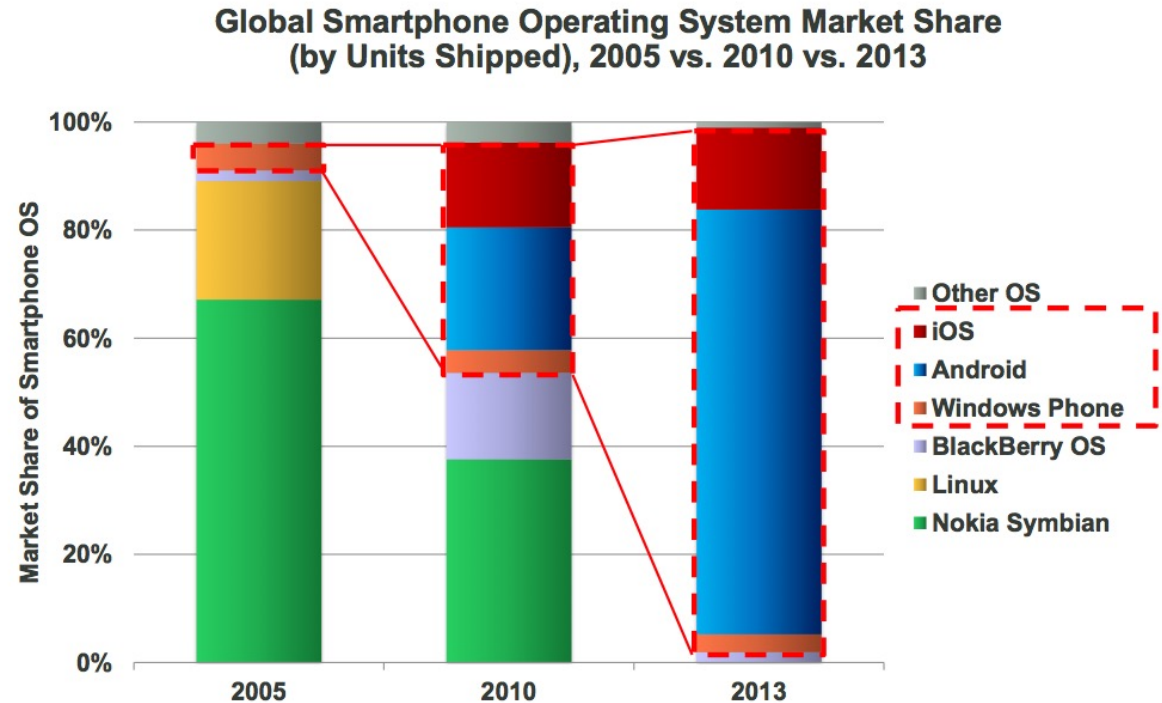
India Internet Users = +28% (2016-June) vs. 40% Y/Y Growth
@ 27% Penetration 355MM Users #2 Behind China

India Internet Users (MM) & Penetration (%), Monthly Active*,
Mid-Year (June) 2009 – 2016E



Global Smartphone Operating Systems 'Made in USA'... 97% Share from 5% Eight Years Ago

Examining smartphone operating systems, over the past seven years, iOS and Android have made major gains with Nokia disappearing and Linux a very small piece of the pie



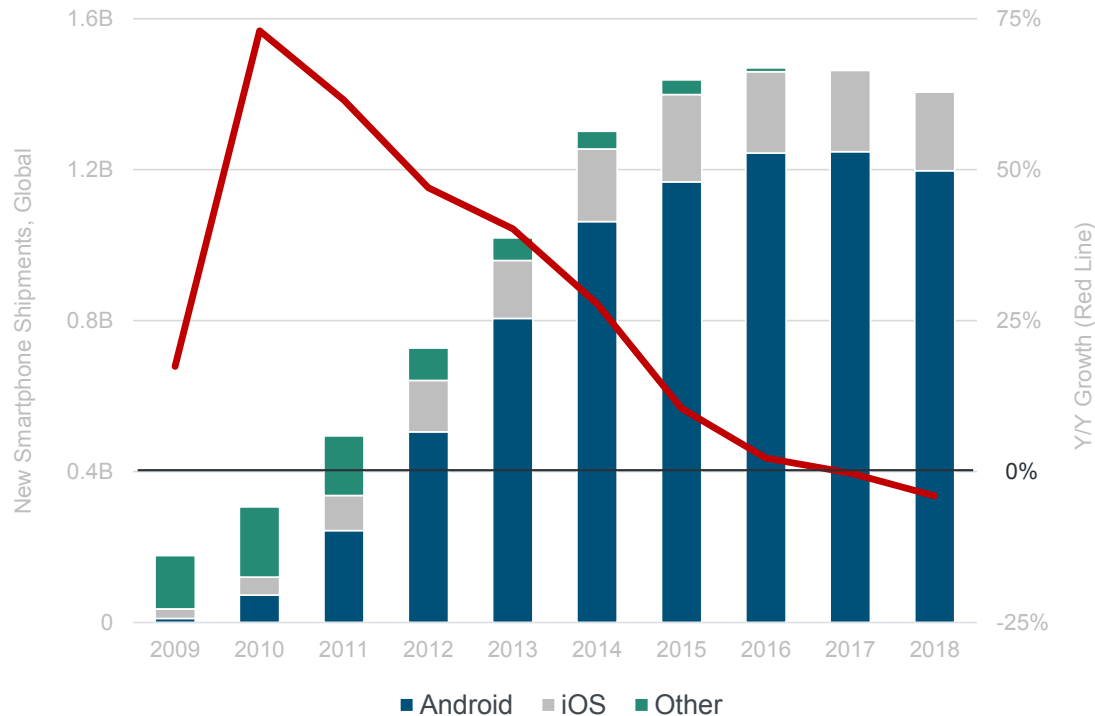
@KPCB

Source: 2005 & 2010 data per Gartner, 2013 data per IDC.

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Global New Smartphone Unit Shipments = Declining -4% vs. 0% Y/Y

New Smartphone Unit Shipments vs. Y/Y Growth



Data Volume = Extraordinary Growth... ~13% Structured / Tagged & Rising Rapidly

There has been exponential growth
in online information;

1 Zettabyte = 1,024 Exabytes

1 Exabyte = 1,024 Petabytes

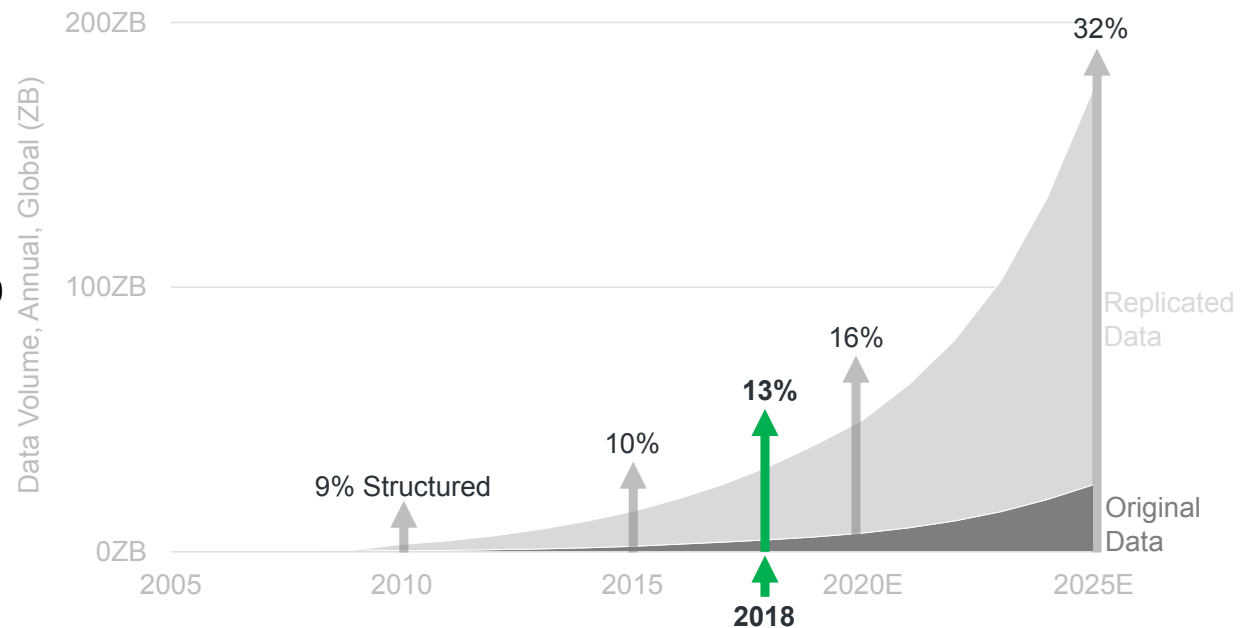
1 Petabyte = 1,024 Terabytes

1 Terabyte = 1,024 Gigabytes

or

1 Zettabyte = 1,000,000,000,000
gigabytes

New Data Captured / Created / Replicated, per IDC

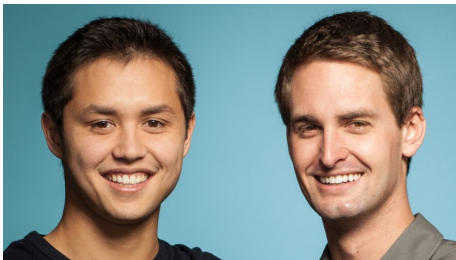


Photos Alone = 1.8B+ Uploaded & Shared Per Day... Growth Remains Robust as New Real-Time Platforms Emerge

500 million photos are uploaded every day and that number is doubling every year

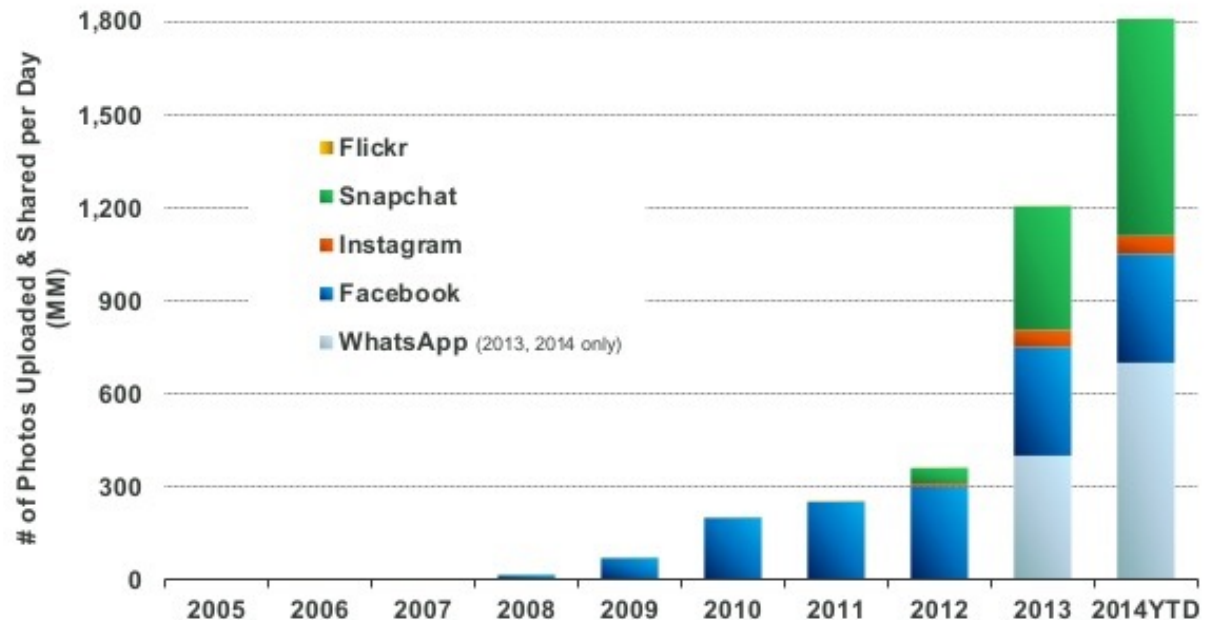
Instagram was recently (2010) purchased by Facebook for \$1 billion

Snapchat is a photo messaging application developed by two Stanford students (IPO March 2017, \$17B valuation);



bobby Murphy - Evan Spiegel

Daily Number of Photos Uploaded & Shared on Select Platforms, 2005 – 2014YTD



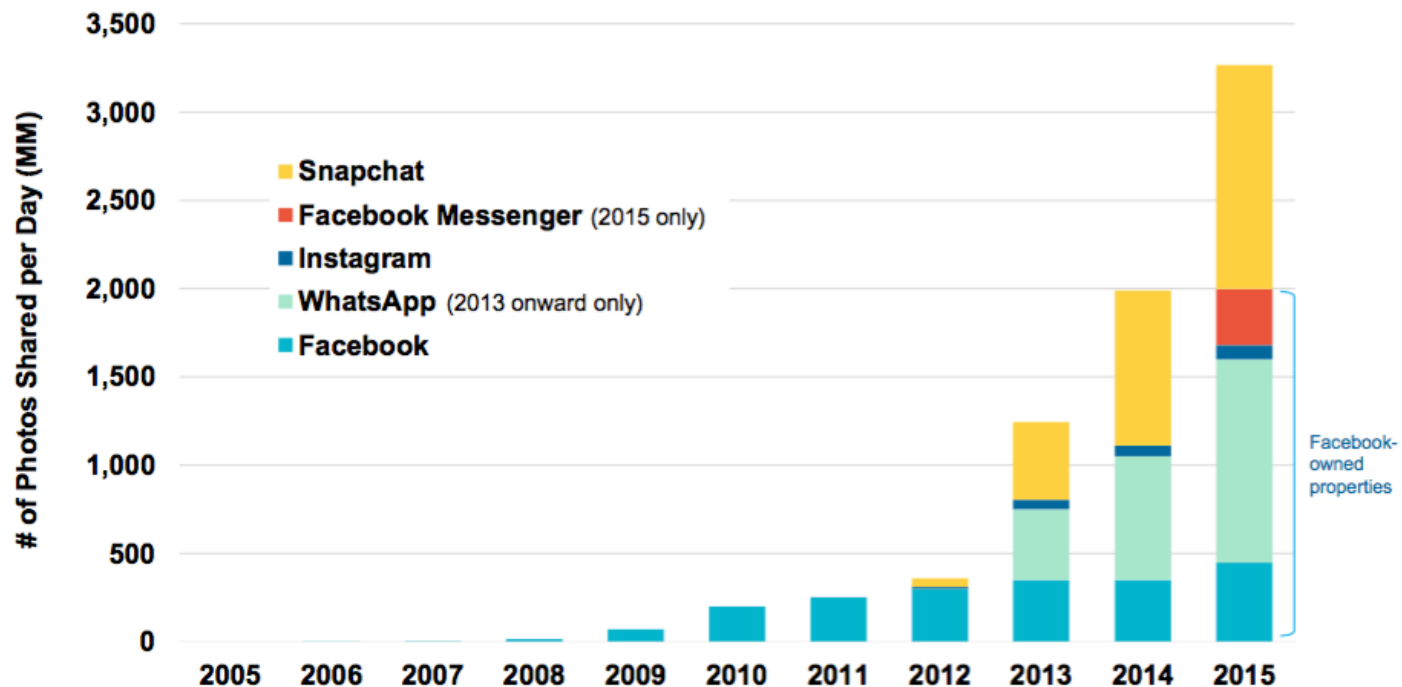
@KPCB

Source: KPCB estimates based on publicly disclosed company data, 2014 YTD data per latest as of 5/14.

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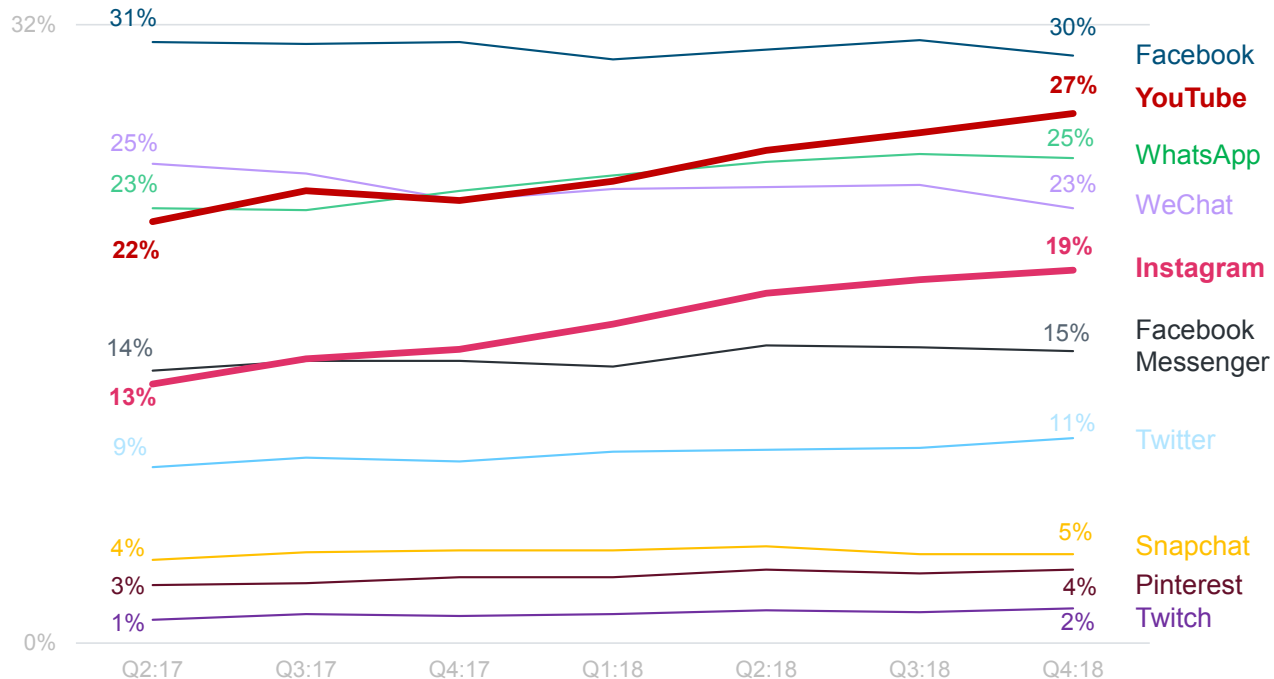
Image Growth Remains Strong

Daily Number of Photos Shared on Select Platforms, Global, 2005 – 2015



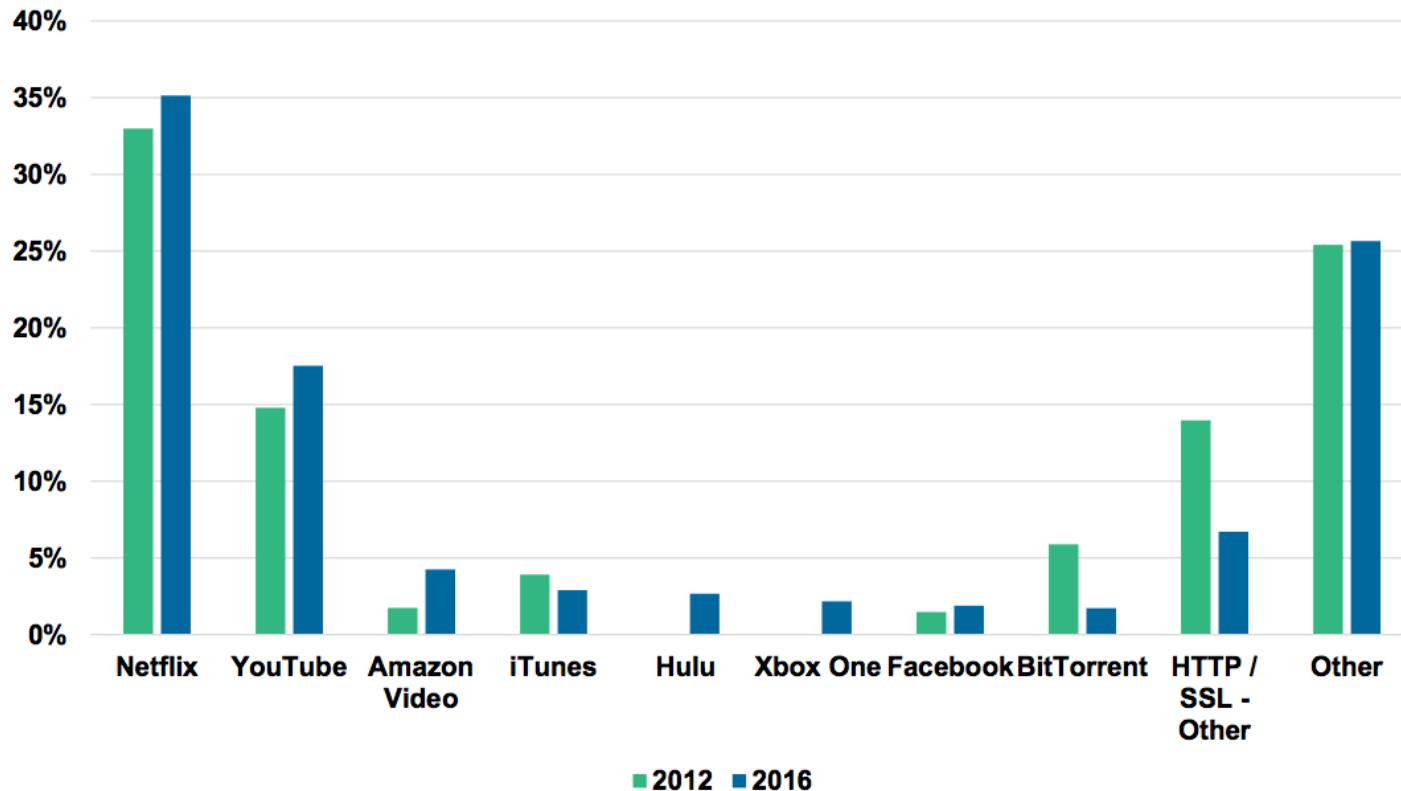
Online Platform Time = YouTube + Instagram Gaining Most

% Internet Users Using Select Platforms > 1x per Day, Global*



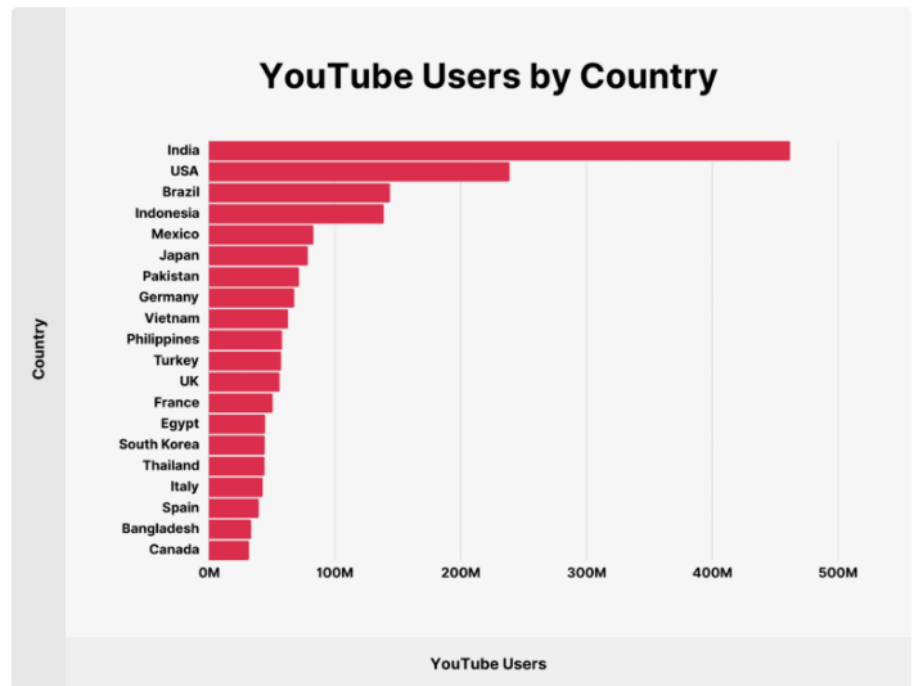
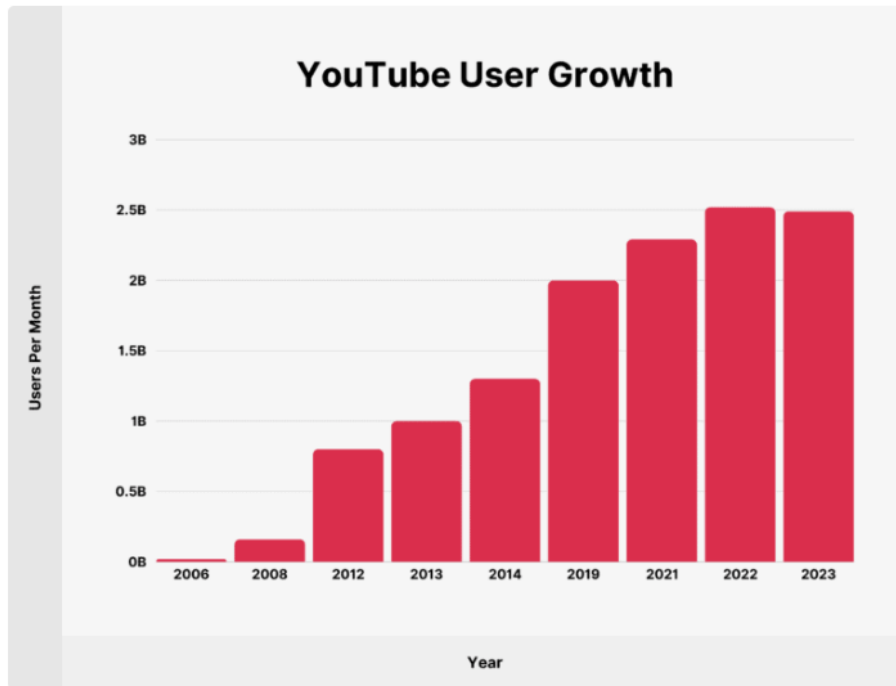
Netflix / YouTube = *Fixed-Access Video* Traffic Share Leaders

Share of Downstream Video Traffic (%), North America, 2H 2016



YouTube Users 2024

YouTube currently has 2.49 billion users, half of the global online population

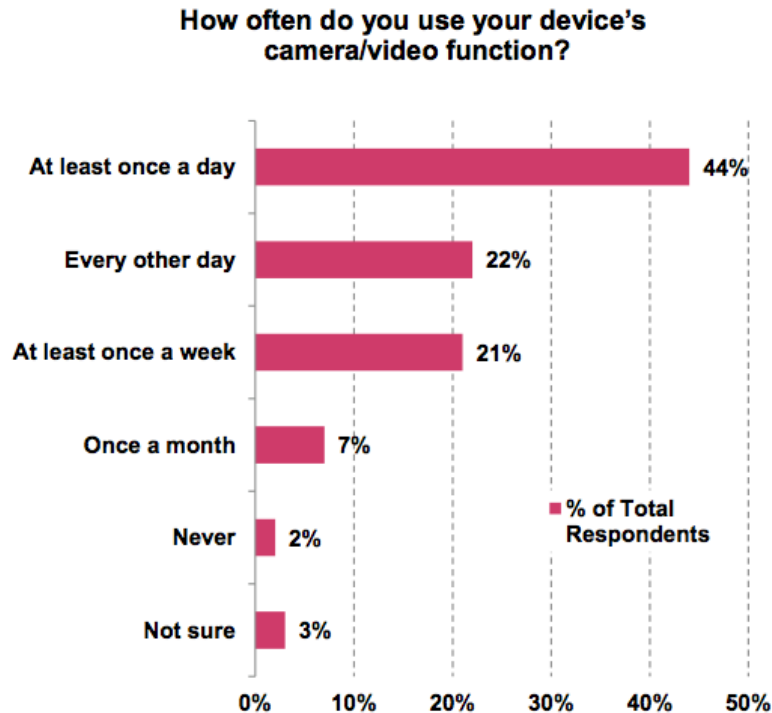


Source; <https://backlinko.com/youtube-users>

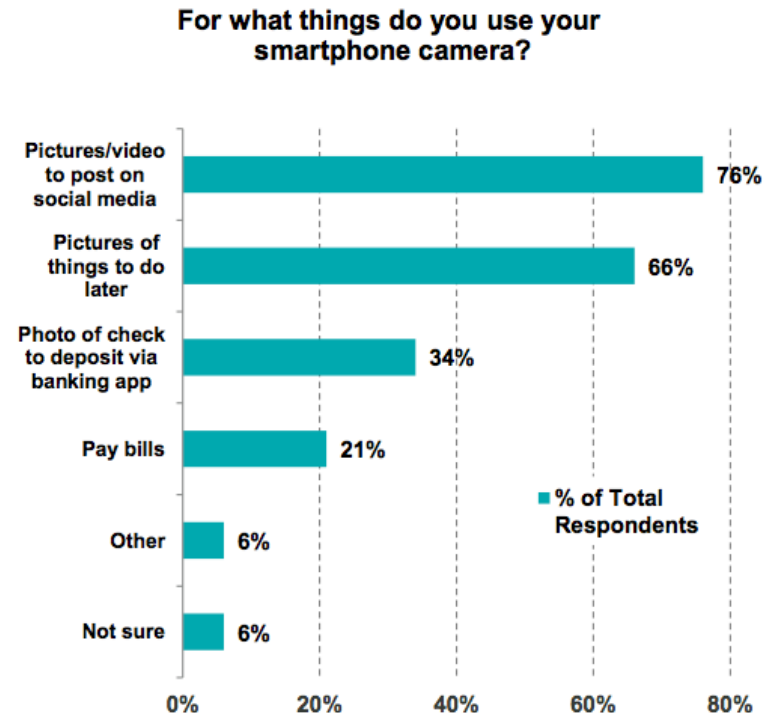
Millennials Love Their Smartphone Cameras...

44% Use Camera / Video Function Daily...76% Post on Social Media

Millennial Smartphone Camera Usage*, USA, 2014



Millennial Smartphone Camera Use Cases, USA, 2014



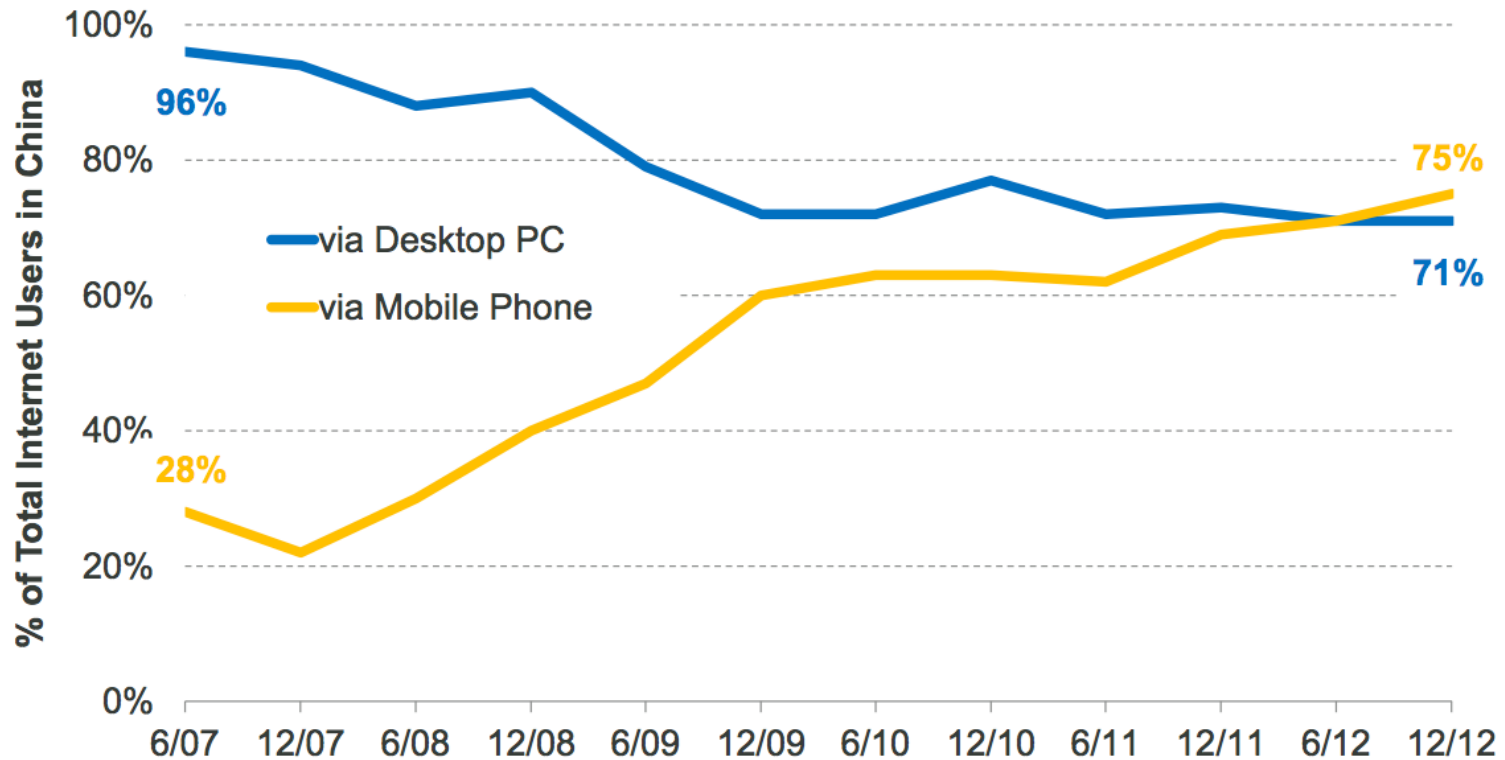
Source: Zogby Analytics.
*18-24 year olds.

Note: Zogby Analytics was commissioned by Mitek Systems, Inc. to conduct an online survey of 1,019 millennials who have a smartphone. For the purposes of this survey, "millennials" are defined as adults between the ages of 18-34. All interviews were completed May 30 through June 6, 2014.

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China – Mobile Internet Access Surpassed PC, Q2:12

**% of Chinese Internet Users Accessing the Web
via Desktop PCs vs. via Mobile Phones, 6/07 – 12/12**

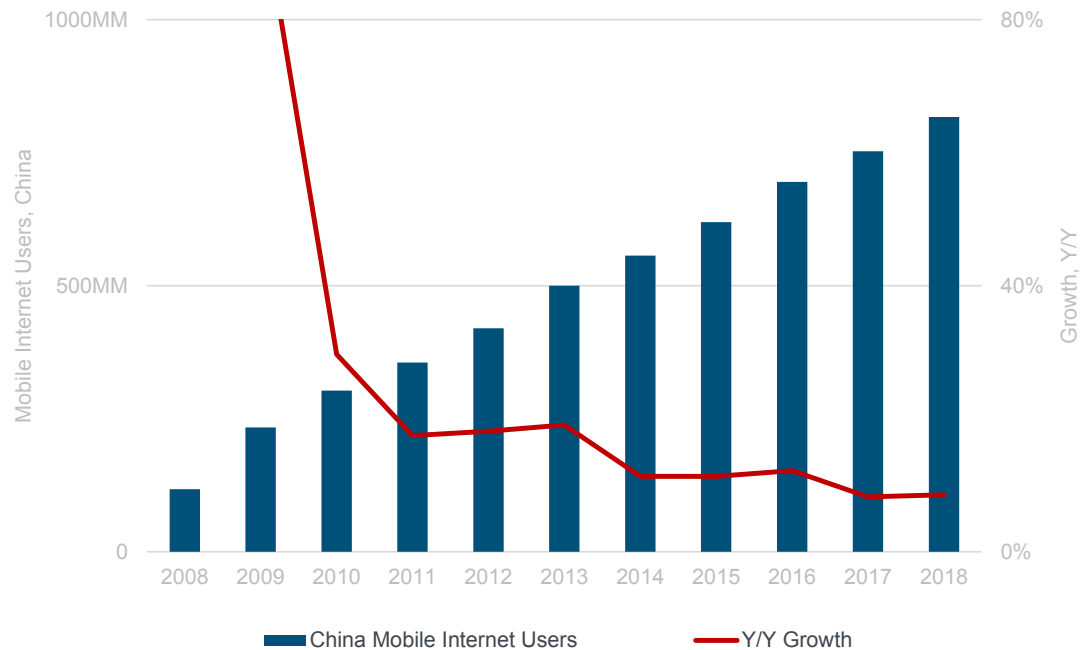


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Source: CNNIC, 1/13. 33

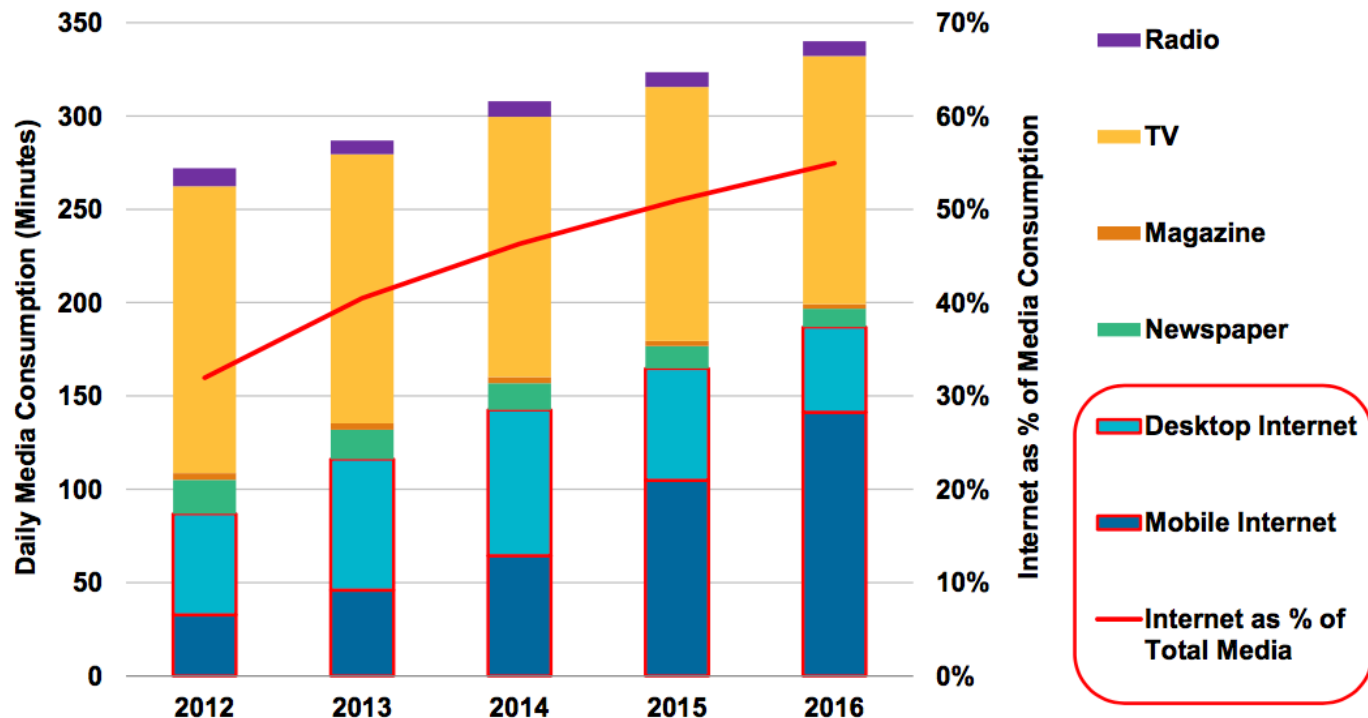
China Mobile Internet Users =
817MM...+9% vs. +8% Y/Y

China Mobile Internet Users vs. Y/Y Growth



China Media = Internet @ 55% of Time Spent Mobile > TV (2016)

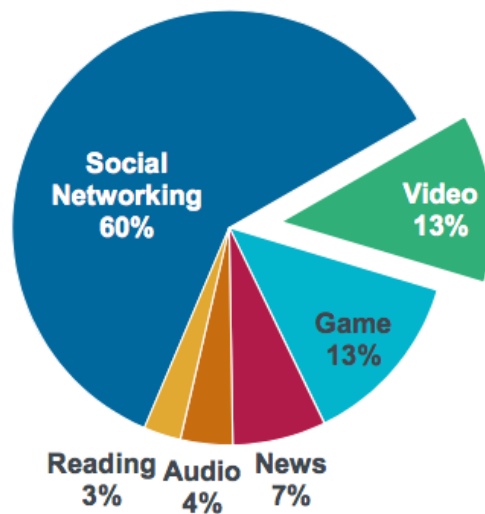
Average Daily Media Consumption Minutes by Medium, China, 2012 - 2016



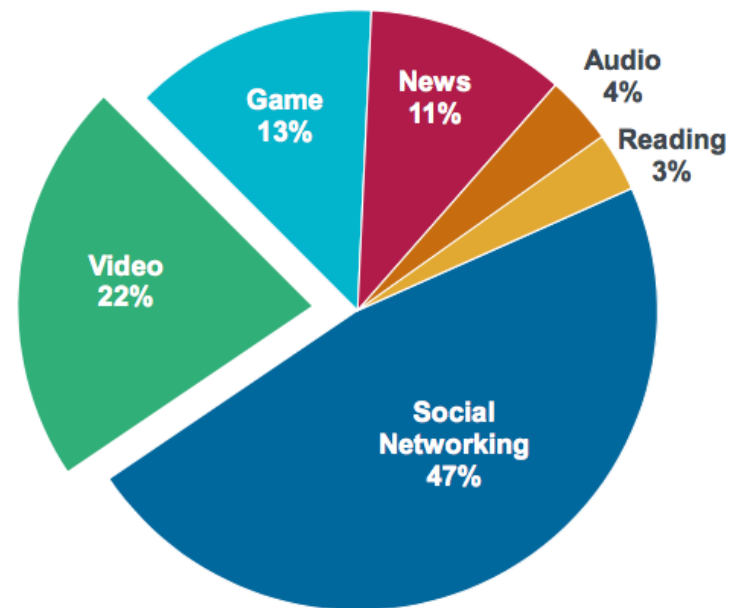
China Mobile Media / Entertainment Time Spent = +22% Y/Y...Mobile Video Growing Fastest

China Mobile Media / Entertainment Daily Time Spent

March 2016
2.0B Hours

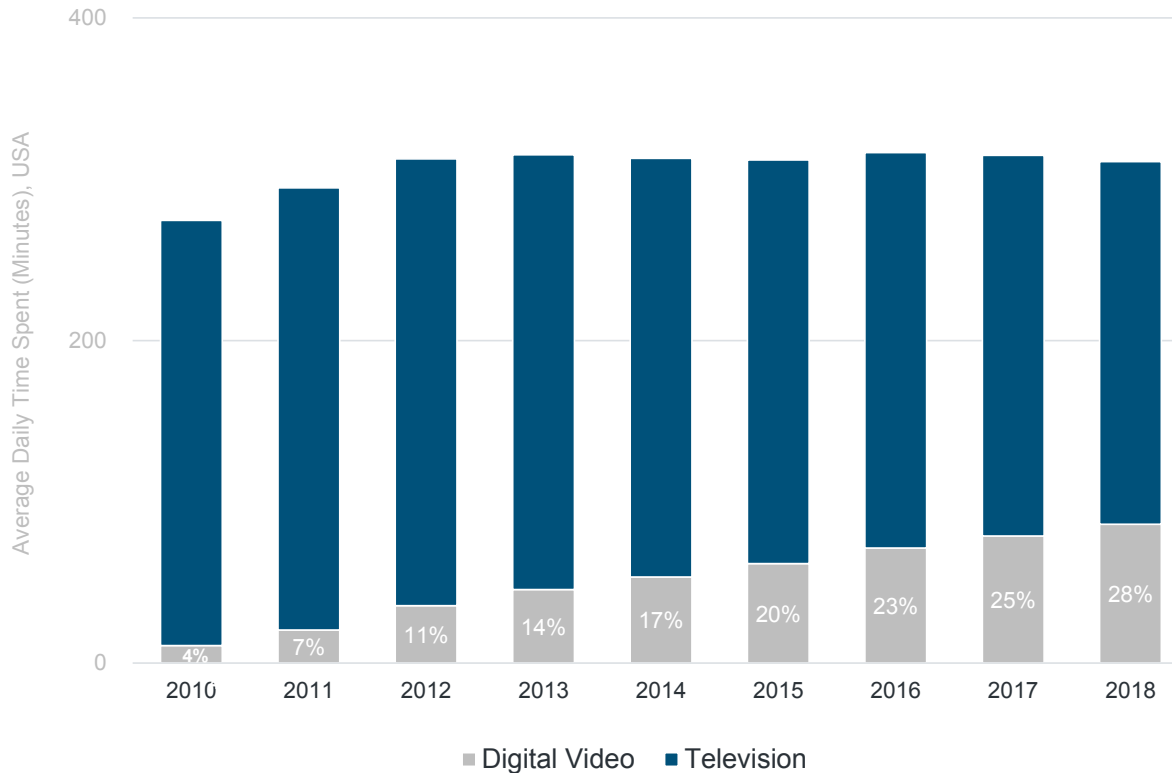


March 2018
3.2B Hours, +22% Y/Y



Video Time =
Digital +2x in Five Years @ 28% of Total (vs. TV)

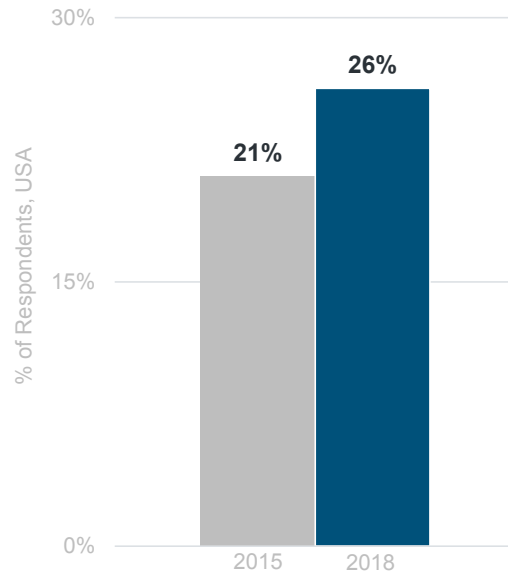
Video Watching Daily Minutes – Digital vs. TV, USA



Adults 'Almost Constantly' Online = 26% vs. 21% Three Years Ago

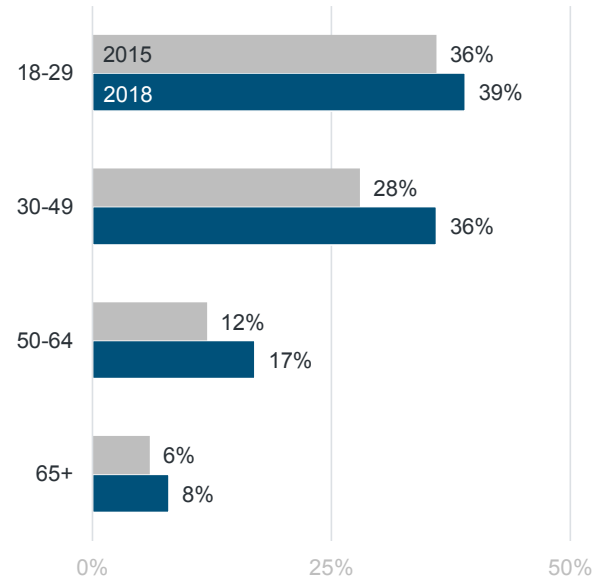
% of Adults Online 'Almost Constantly'

Overall



Pew Survey (USA)

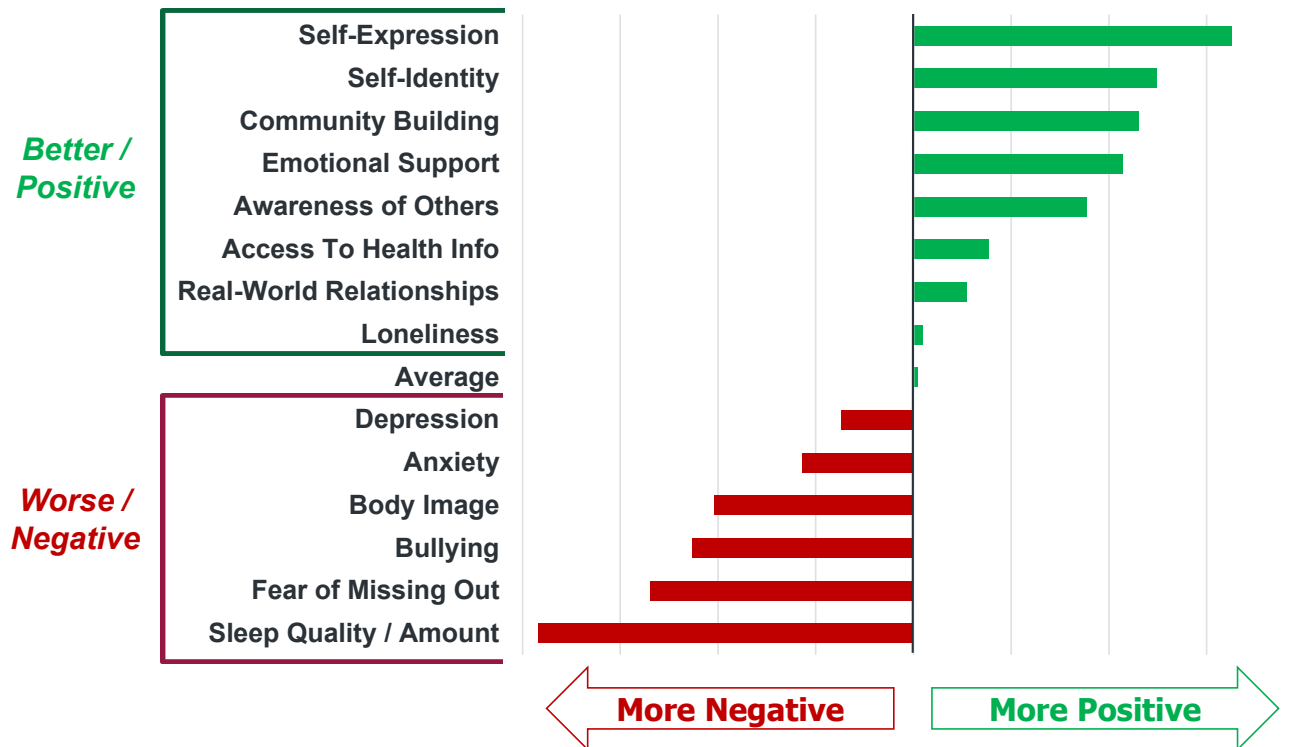
By Age Group



% of Respondents in Pew Survey, USA

Social Media = Positive & Negative

*Do Social Media Platforms You Use Make These
Health-Related Factors Better or Worse?*

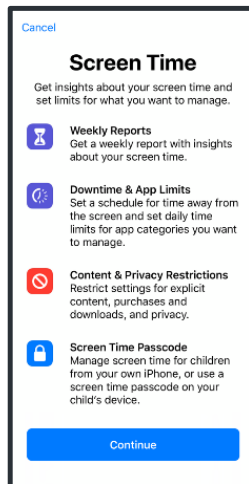


Digital Media = Businesses Taking Action to Help Users Monitor Usage

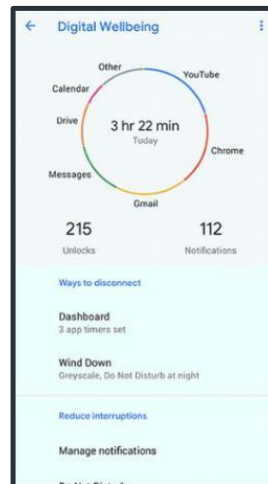
2018

Major Platforms Launched Wellness / Time Tracking Features

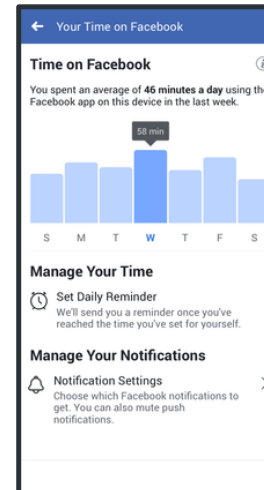
Apple Screen Time



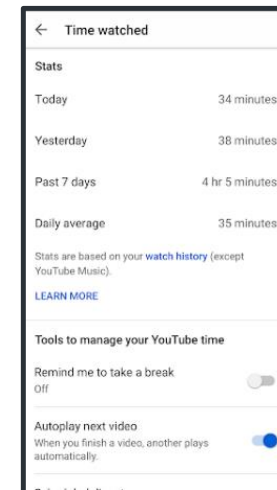
Google Digital Wellbeing



Facebook Your Time on Facebook



YouTube Time Watched



Technology Cycles – Still Early Cycle on Smartphones + Tablets, Now Wearables Coming on Strong, Faster than Typical 10-Year Cycle

Technology Cycles Have Tended to Last Ten Years

*Mainframe
Computing
1960s*



*Mini
Computing
1970s*



*Personal
Computing
1980s*



*Desktop Internet
Computing
1990s*



*Mobile Internet
Computing
2000s*



*Wearable /
Everywhere
Computing
2014+*



Others?

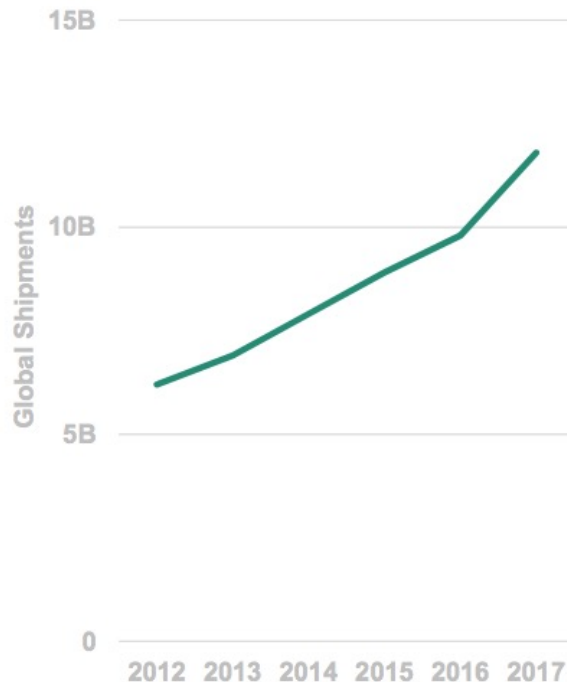
KPCB

Image Source: Computersciencelab.com, Wikipedia, IBM, Apple, Google, NTT docomo, Google, Jawbone, Pebble.

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...Data Gathering + Sharing + Optimization (2006 →) = Enabled by Sensor Pervasiveness...

MEMS Sensor / Actuator Shipments



Sensors + Data = In More Places

Visual Navigation
Google Maps



Shared Transportation
Mobike



Home Temperature
Nest



Predictive Maintenance
Samsara



Fitness Tracking
Motiv

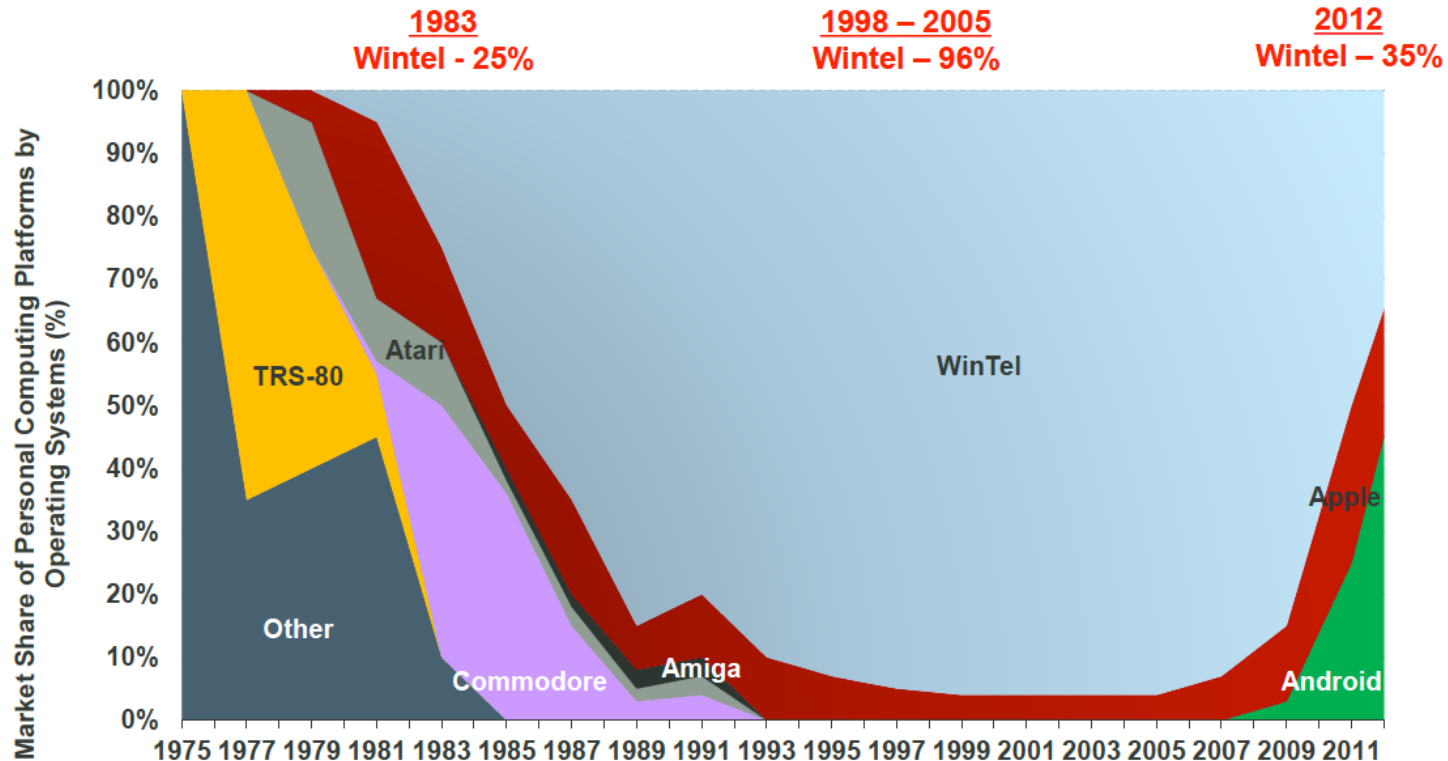


Precision Cooking
Joule



Re-Imagination of Computing Operating Systems - iOS + Android = 60% Share vs. 35% for Windows

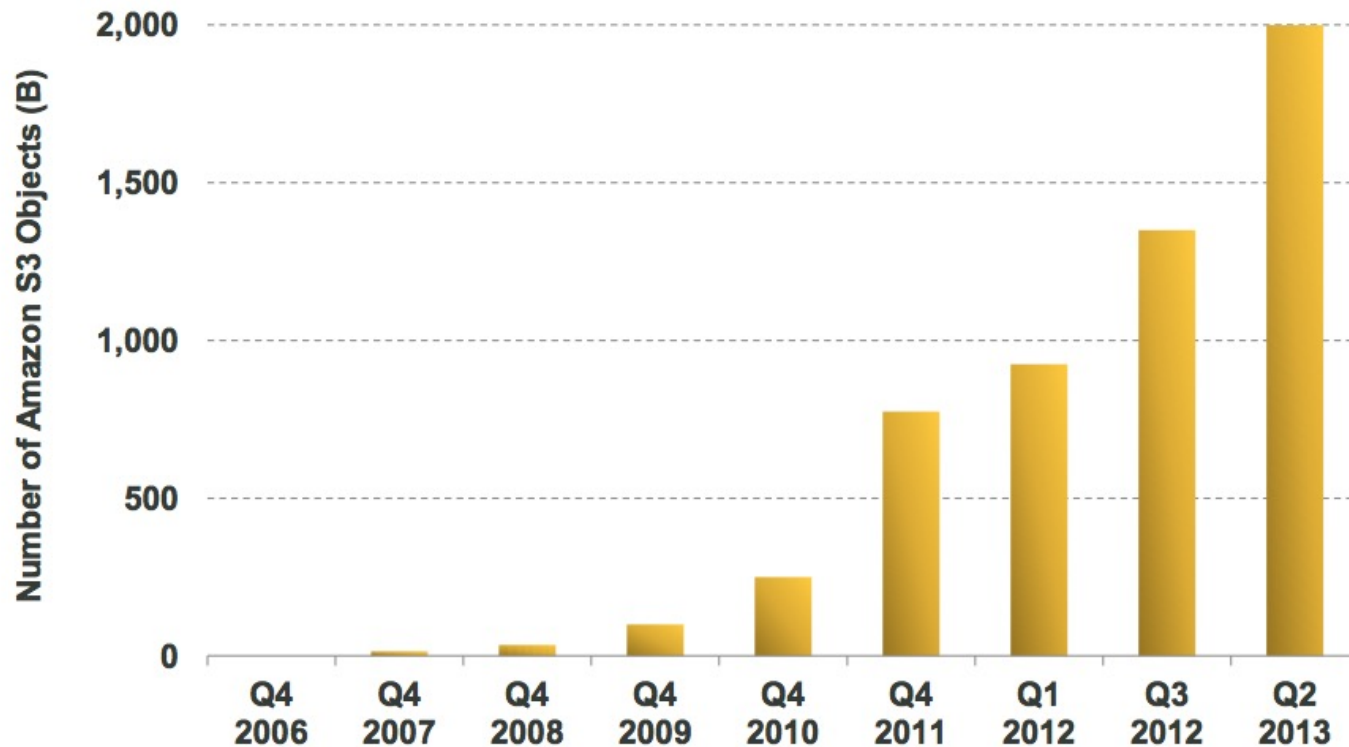
Global Market Share of Personal Computing Platforms by Operating System Shipments, 1975 – 2012



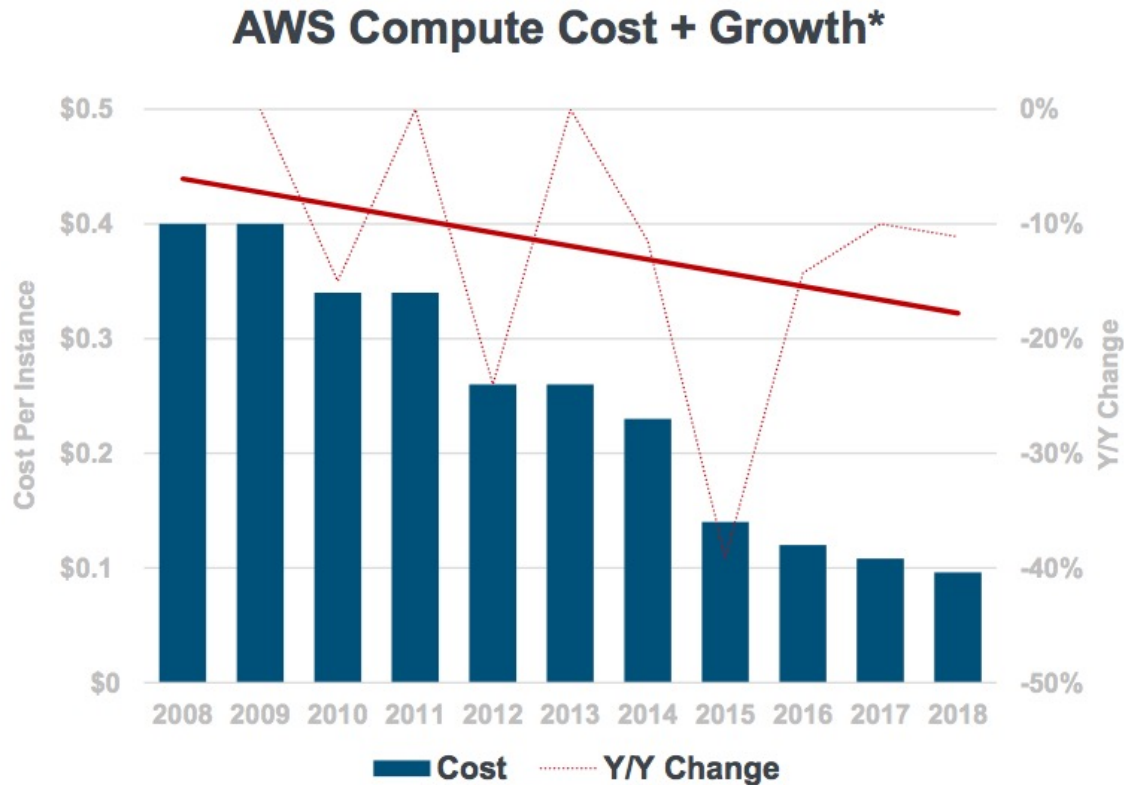
...While The Cloud Rises

Amazon Web Services (AWS) Leading Cloud Charge...

Objects Stored in Amazon S3* (B)

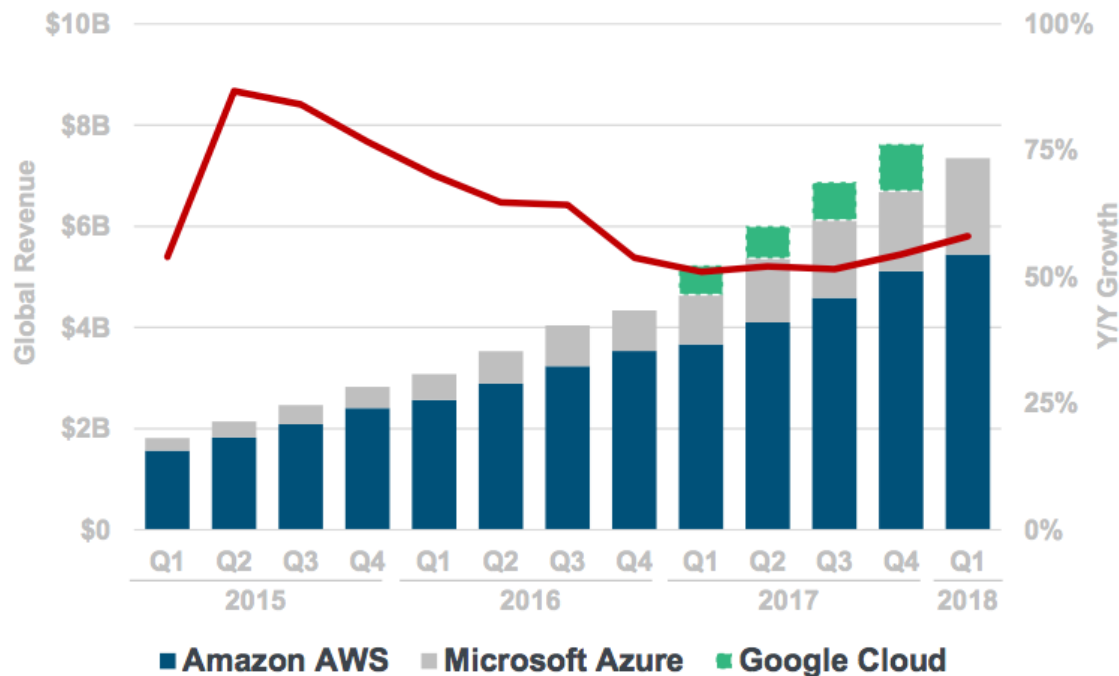


...Computing Big Bangs Volume Effects = Cloud Compute Cost Declines Continue -11% vs. -10% Y/Y...



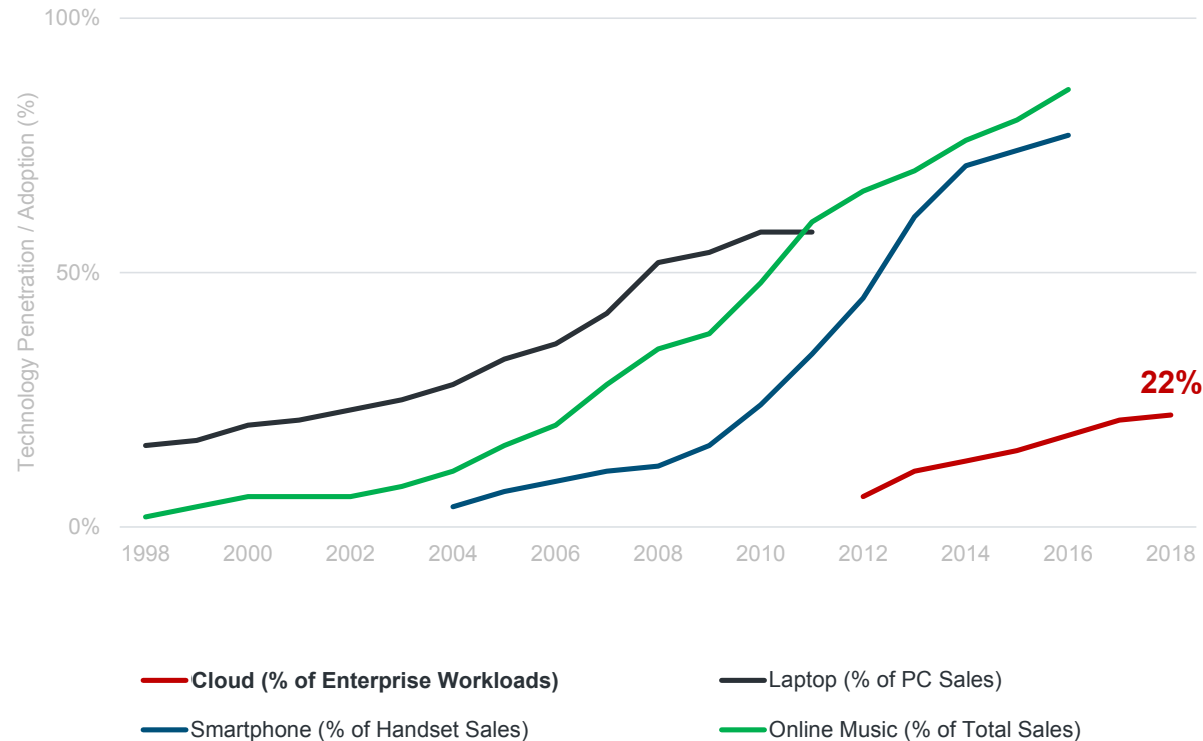
...Computing Big Bangs Volume Effects = Cloud Revenue Re-Accelerating +58% vs. +54% Q/Q

Cloud Service Revenue – Amazon + Microsoft + Google



...Cloud Deployment =
22% of Workloads +2x vs. Five Years Ago

Technology Adoption Rates, Global per Morgan Stanley



Cloud Evolution / Tools = Paving Way for Innovation Across Infrastructure Landscape



New Methods of Software Delivery =

APIs / Browser Extensions creating new wave of capabilities (+ companies) for both companies and end users



Containers / Microservices =

Simplify software development process / improve consistency between testing & production environments / reduce complexity of managing & updating apps due to modular approach



Elastic Analytical Databases =

Likes of Google BigQuery / Snowflake / AWS Redshift Spectrum nearly infinitely scalable / usage based + have minimal maintenance requirements

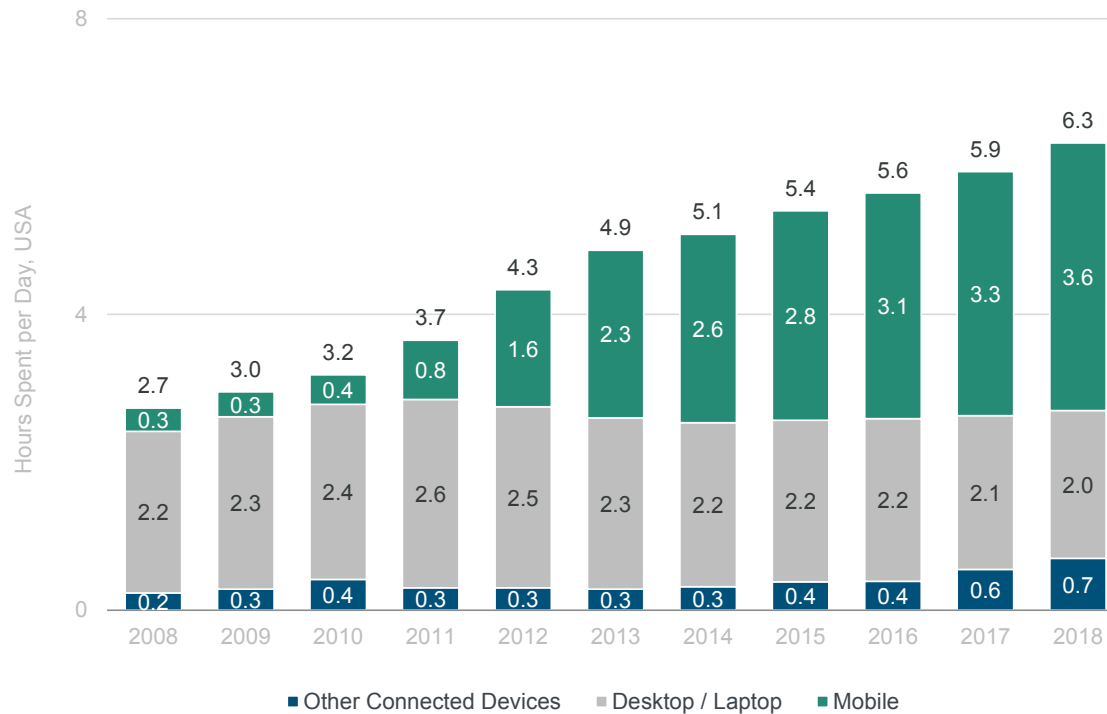


Edge Computing =

Pushing compute away from centralized nodes & closer to sources of data addresses many IT challenges when running data-centric workloads in cloud – reduces latency / can have security + compliance benefits

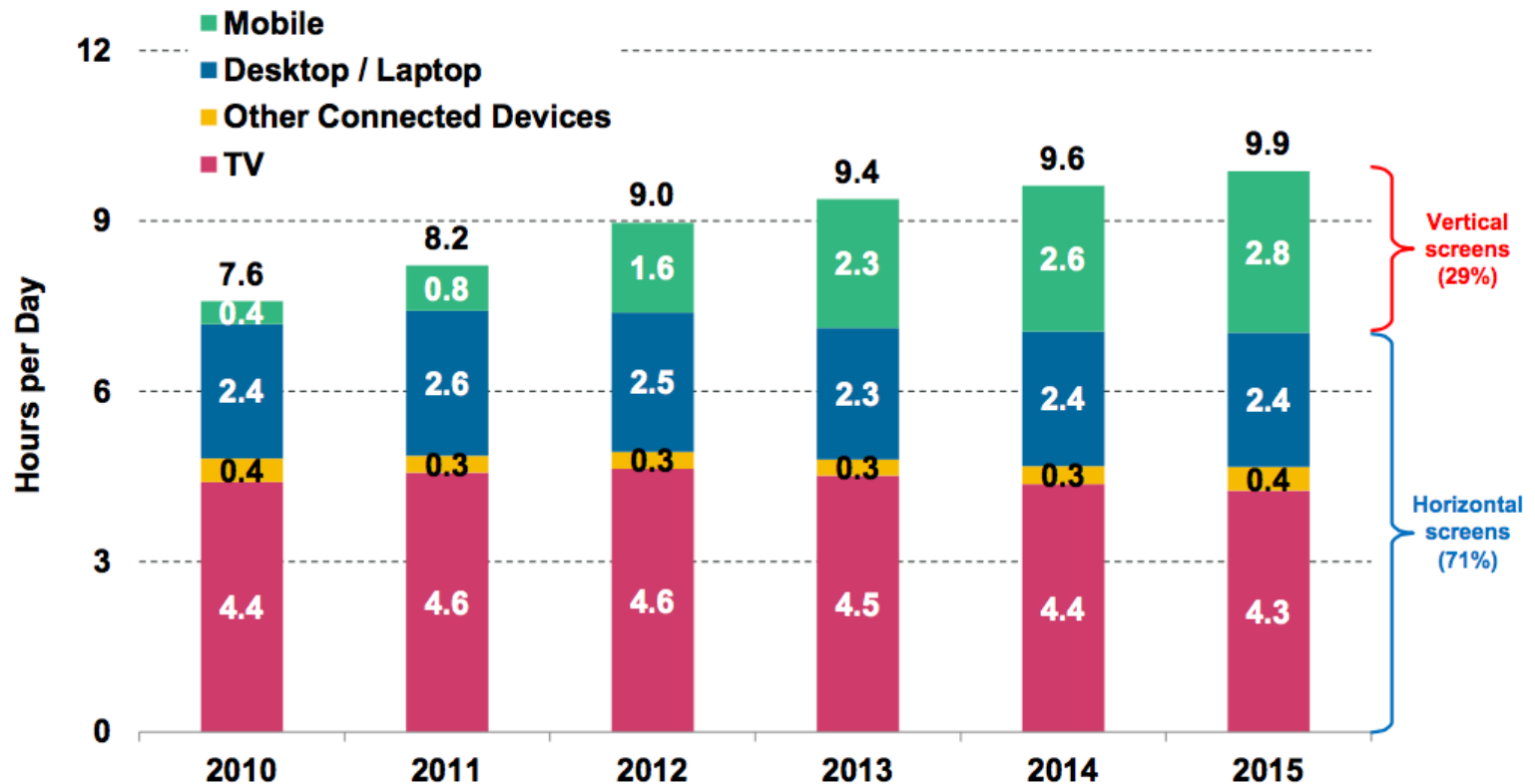
Digital Media Usage = Accelerating +7% vs. +5% Y/Y

Daily Hours Spent with Digital Media per Adult User, USA



...Vertical Viewing = 29% of View Time (Multi-Platform) vs. 5% Five Years Ago, USA...

Time Spent on Screens by Orientation (Hours / Day), USA, 2010 – 2015



Source: eMarketer 4/15, Coatue analysis. Note: Other connected devices include OTT and game consoles. Mobile includes smartphone and tablet. Usage includes both home and work. Ages 18+; time spent with each medium includes all time spent with that medium, regardless of multitasking; for example, 1 hour of multitasking on desktop/laptop while watching TV is counted as 1 hour for TV and 1 hour for desktop/laptop.

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Messaging Apps = Top Global Apps in Usage + Sessions

6+ of Top 10
most used apps
globally =
Messaging Apps

Top Apps by Usage

Rank	App
①	 Facebook
②	 WhatsApp
③	 Messenger
④	 Instagram
⑤	 LINE
⑥	 Viber
⑦	 KakaoTalk
⑧	 Clash of Clans
⑨	 WeChat
⑩	 Twitter

Top Apps By Number of Sessions

Rank	App	Sessions
①	 KakaoTalk	55
②	 WhatsApp	37
③	 WeChat	29
④	 VK	29
⑤	 LINE	26
⑥	 Viber	20
⑦	 Facebook	20
⑧	 Clash of Clans	16
⑨	 Instagram	12
⑩	 Messenger	8

Messaging
Apps →
significant app
sessions



Source: Quettra, Q1:15. Data ranked based on usage.

Quettra analyzes 75MM+ Android users spread out in more than 150 countries, collecting install and usage statistics of every application present on the device. Q1:15 data analyzed three months of data starting from 1/1/15. Data excludes Google apps and other commonly pre-installed apps to remove biases. Only apps with 10K+ installs worldwide and 100+ DAU are counted.

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Messaging = Extensibility Expanding

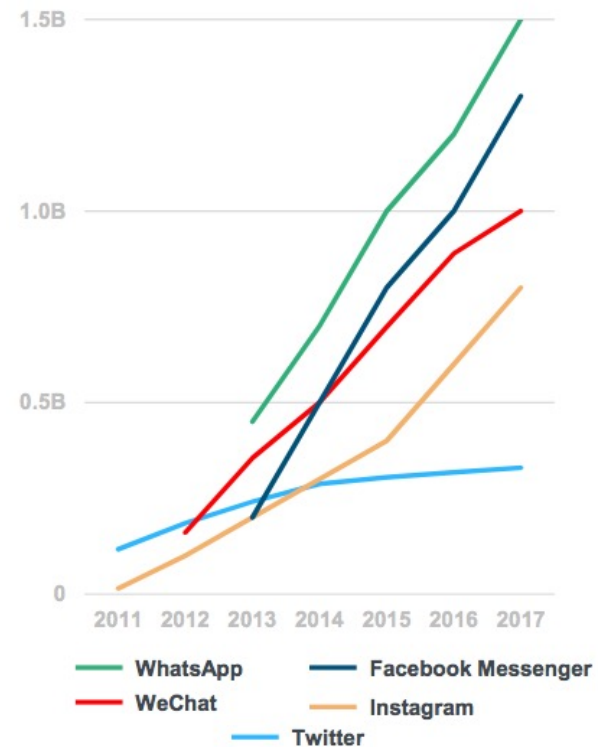
Messaging Tencent (2000 → 2018)



I am on WeChat: marcopapa99

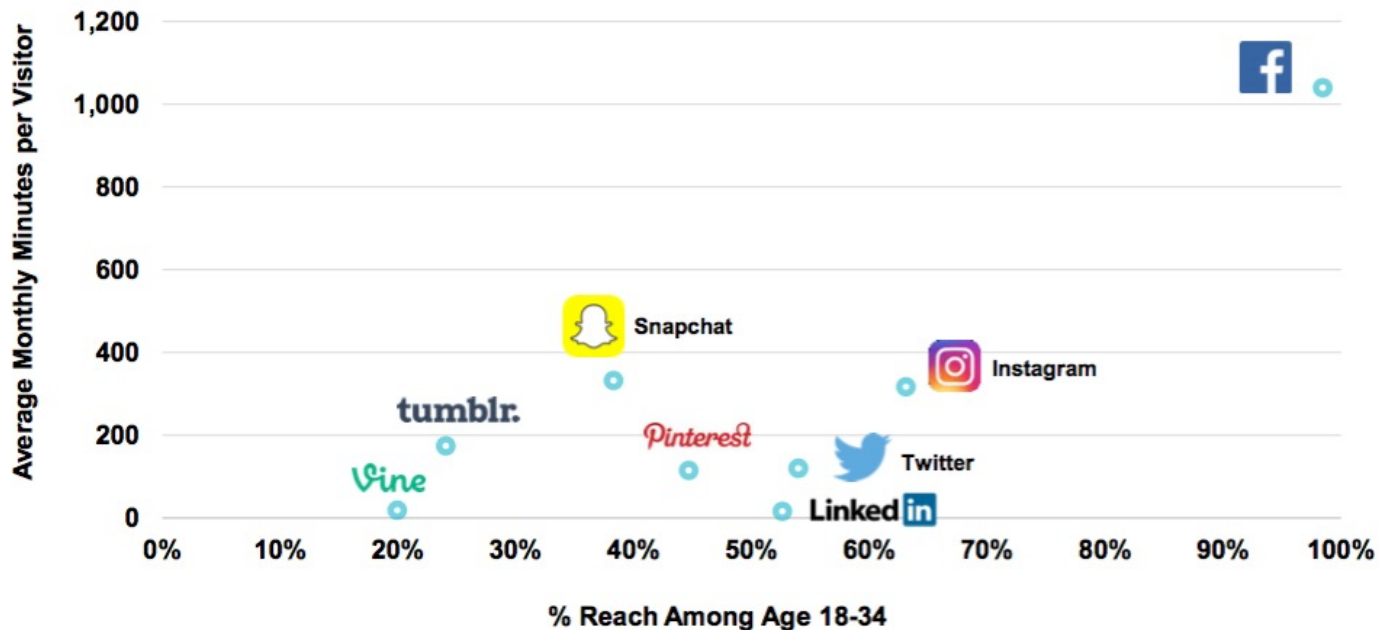
Messenger MAUs

MAU = Monthly Active Users



Millennial Social Network Engagement Leaders = Visual... Facebook / Snapchat / Instagram...

**Age 18-34 Digital Audience Penetration vs.
Engagement of Leading Social Networks, USA, 12/15**



Asia-Based Messaging Leaders = Continue to Expand Uses / Services Beyond Social Messaging

			
Name	KakaoTalk	WeChat	LINE
Launch	March 2010	January 2011	June 2011
Primary Country	Korea	China	Japan
New Services Added 2015 -16*	Banking / Financial Services	Kakao Bank (11/15)	WeBank (1/15)
	Enterprise	✖	Enterprise WeChat (3/16)
	Online-To-Offline (O2O)	Kakao Hairshop (1H:16E) Kakao Driver (1H:16E)	✓
	TV	Kakao TV (6/15)	✓
	Video Calls / Chat	(6/15)	✓
	Taxi Services	Kakao Taxi (3/15)	✓
	Messaging	✓	✓
	Group Messaging	✓	✓
	Voice Calls	Free VoIP calls (2012)	WeChat Phonebook (2014)
	Payments	KakaoPay (2014)	(2013)
Previous Existing Services	Stickers	(2012)	Sticker shop (2013)
	Games	Game Center (2012)	(2014)
	Commerce	Kakao Page (2013)	Delivery support w / Yixun (2013)
	Media	Kakao Topic (2014)	✓
	QR Codes	✓	QR code identity (2012)
	User Stories / Moments	Kakao Story (2012)	WeChat Moments
	Developer Platform	KakaoDevelopers	WeChat API
			Line Pay (2014)

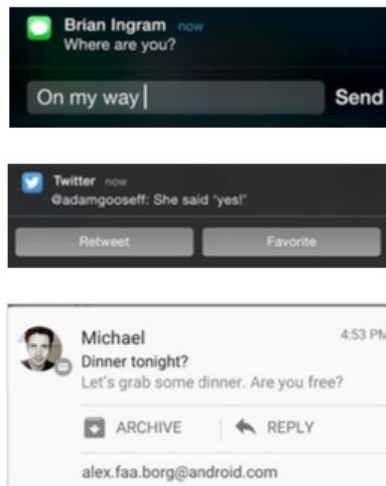
**Average Global Mobile User = ~33 Apps...12 Apps Used Daily...
80% of Time Spent in 3 Apps**

Day in Life of a Mobile User, 2016

	Average # Apps Installed on Device*	Average Number of Apps Used Daily	Average Number of Apps Accounting for 80%+ of App Usage	Time Spent on Phone (per Day)	Most Commonly Used Apps
USA	37	12	3	5 Hours	Facebook Chrome YouTube
Worldwide	33	12	3	4 Hours	Facebook WhatsApp Chrome

Notifications = Growing Rapidly & Increasingly Interactive... Driving New Touch Points with Messaging Platforms + Other Apps

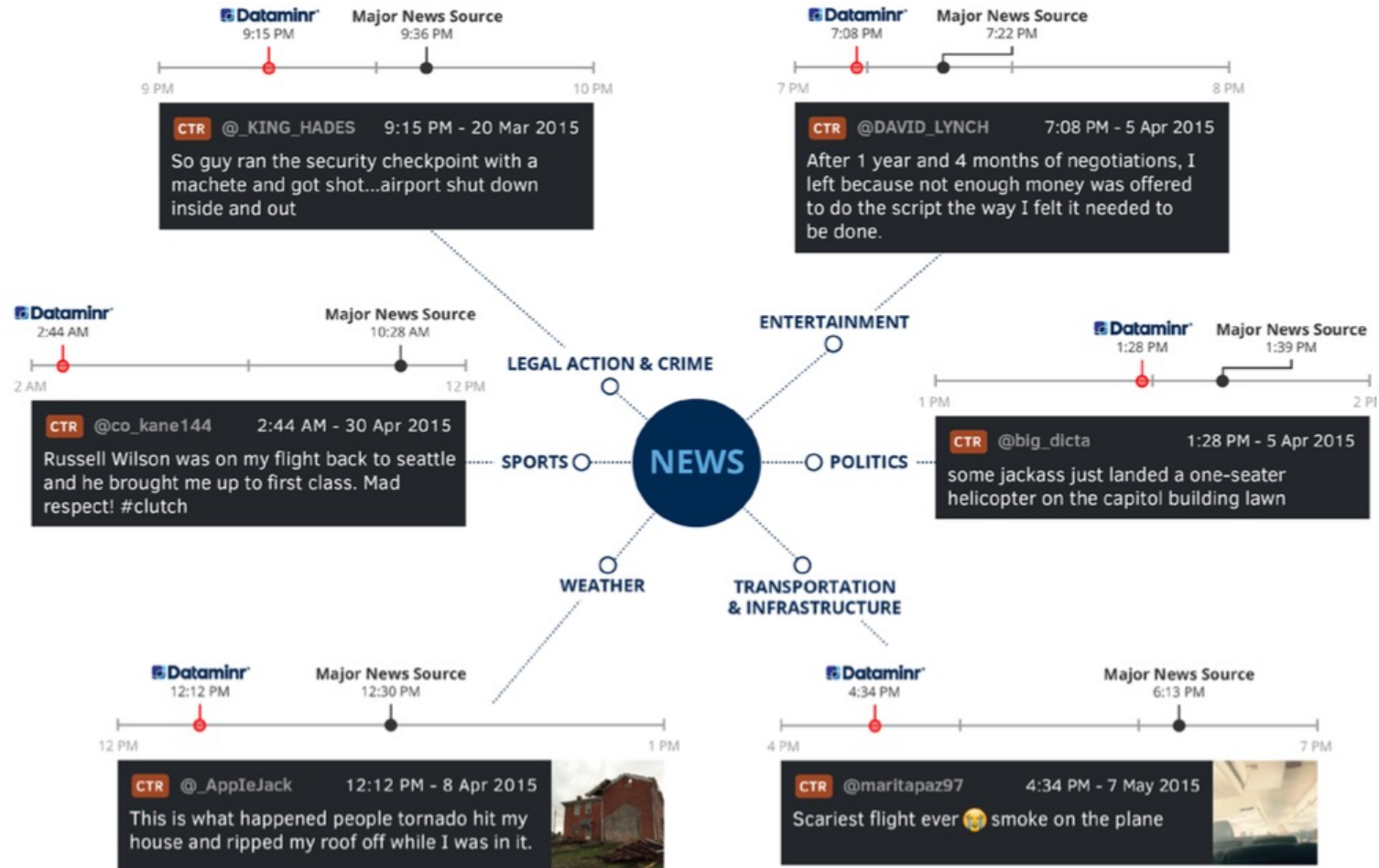
**Direct Interaction
on Notification Panel –**
without users interrupting
what they're doing...



...More Up Close & Personal –
as notifications appear on more
& more mobile devices



Users Increasingly First Source for News via Twitter / Dataminr



@KPCB Source: Dataminr, 5/15.

65

Video Evolution = Accelerating Live (Linear) → On-Demand → Semi-Live → Real-Live

Live (Linear)

Traditional TV
1926

Tune-In or
Miss Out

Mass Concurrent
Audience

Real-Time Buzz



On-Demand

DVR / Streaming
1999

Watch on
Own Terms

Mass Disparate
Audience

Anytime Buzz



Semi-Live

Snapchat Stories
2013

Tune-In Within 24
Hours or Miss Out

Mostly Personal
Audience

Anytime Buzz



Real-Live

Periscope + Facebook Live
2015 / 2016

Tune-In / Watch
on Own Terms

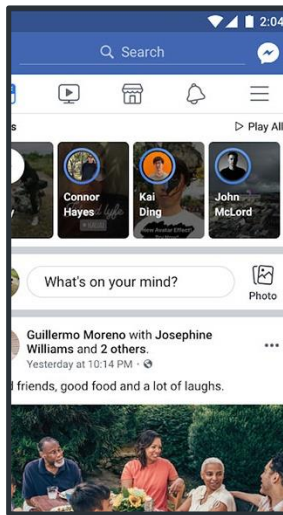
Mass Audience,
yet Personal

Real Time + Anytime Buzz

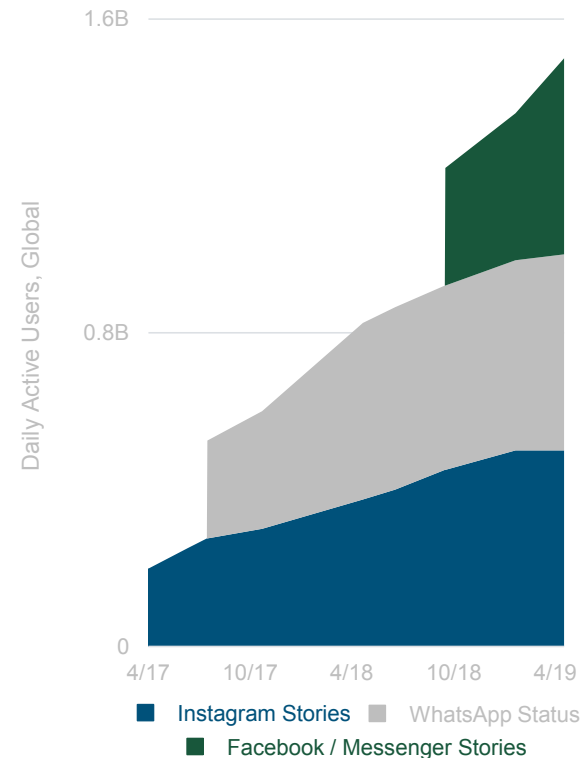


Video Time (Short-Form – Facebook Platform) = 1.5B DAUs + ~2x in One Year

Facebook Stories

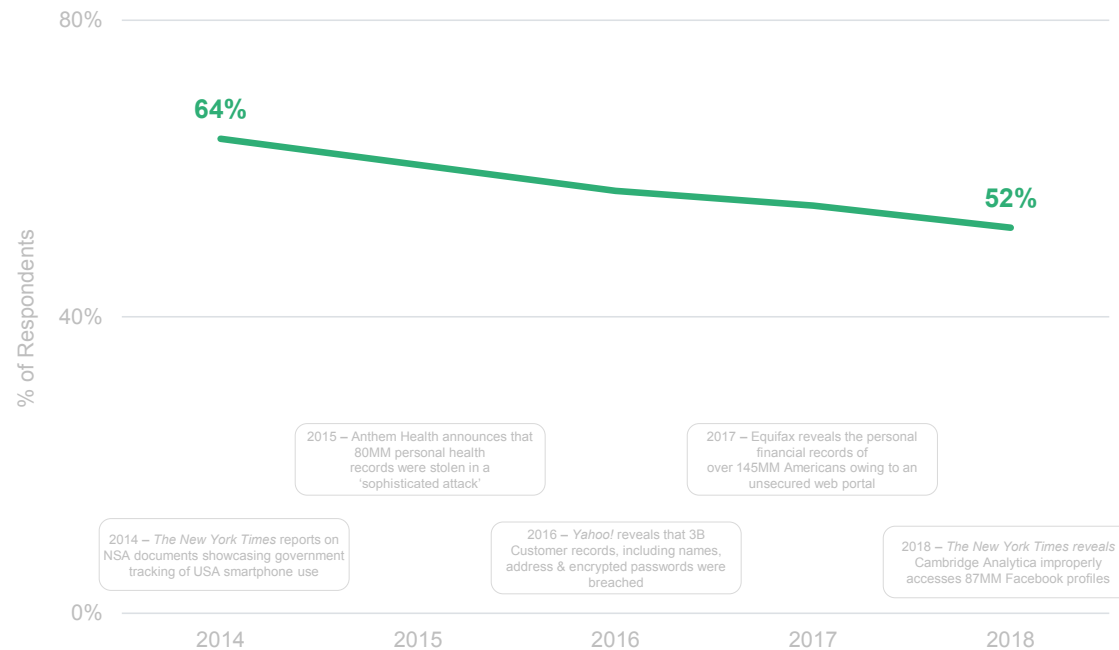


Daily Active Users, Global*

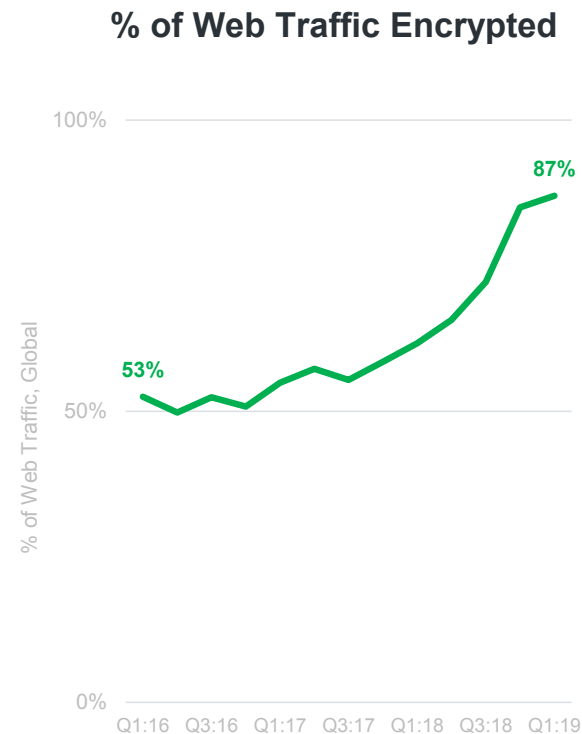
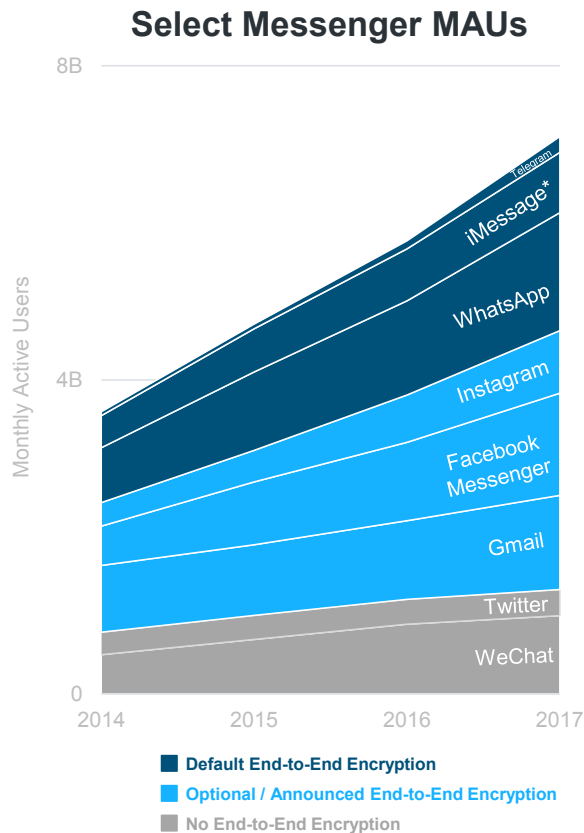


Digital Media = Privacy Concerns High But Moderating

People More Concerned About Internet Privacy vs. One Year Ago, Global

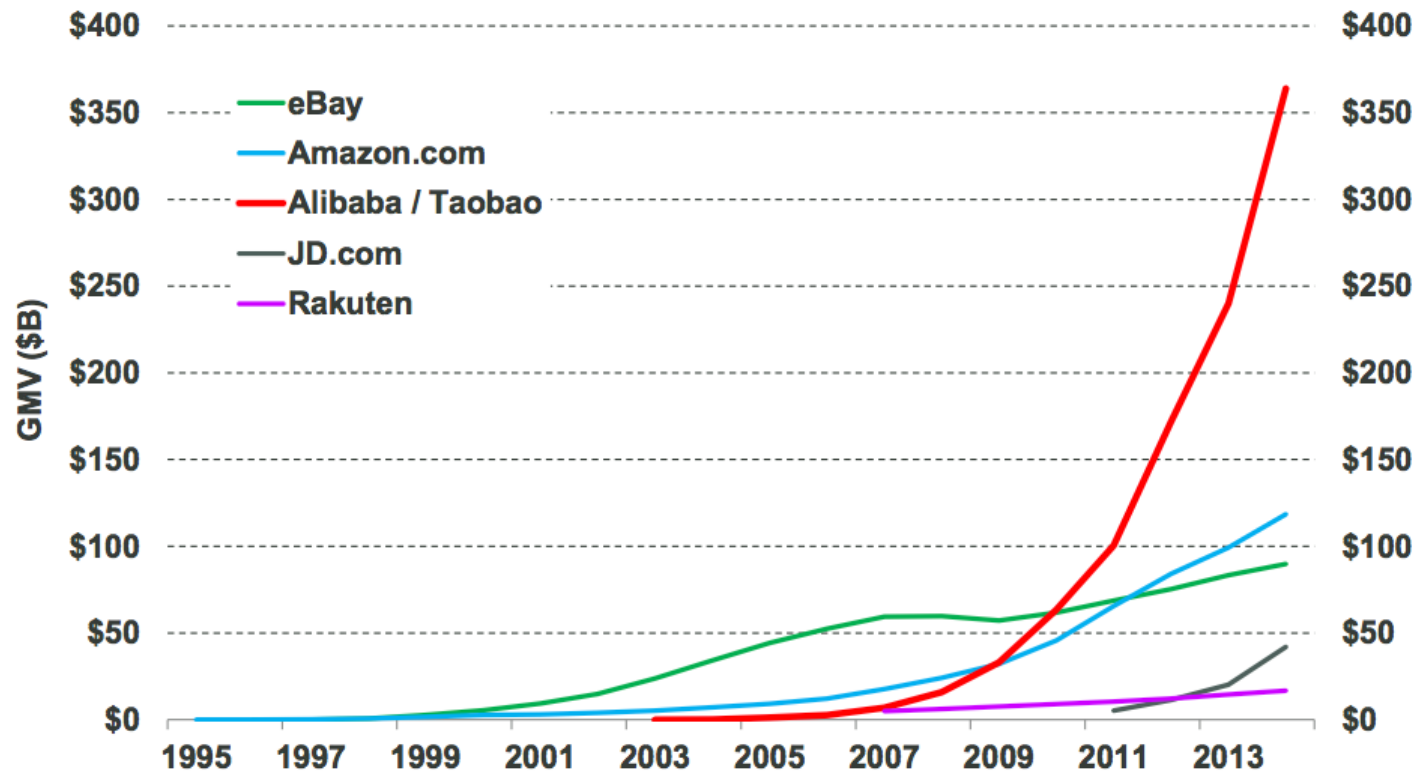


Digital Media = Encrypted Messaging / Traffic Rising Rapidly



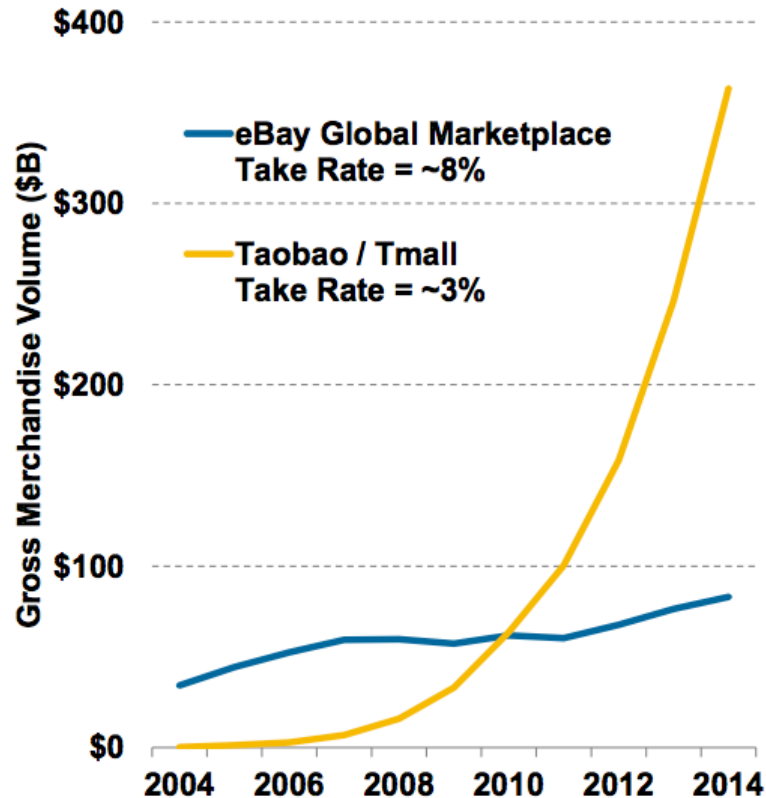
1st Generation 'Online Platforms / Marketplaces for Products Rising = Optimized for Desktop Internet + Traditional Shipping Delivery

Gross Merchandise Value (GMV), 2014...
Measured by Top 5 Global Public E-Commerce Companies

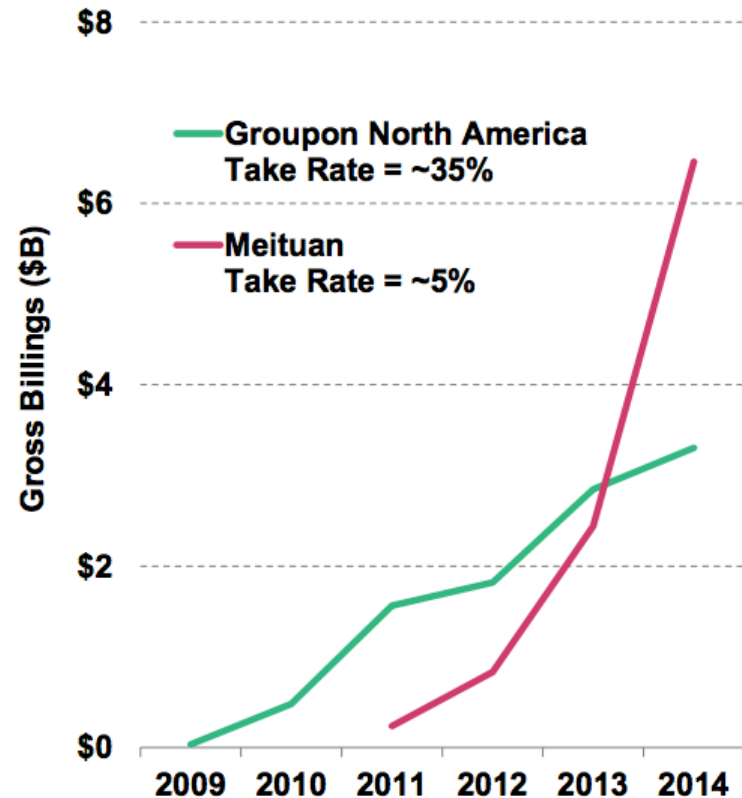


China E-Commerce = Low Take Rates* Helped China Marketplace Leaders Pass USA Peers

**Gross Merchandise Value, 2004 – 2014
eBay vs. Alibaba (Taobao / Tmall)**



**Gross Billings, 2009 – 2014
Groupon N. America vs. Meituan**



@KPCB

Source: Meituan gross billings data are estimates by Tuan800.com, eBay, Groupon, Alibaba GMV data per company.
 Note: Take rate defined as net revenue divided by gross merchandise value or gross billings. eBay marketplace take rate excludes PayPal (~3%), eBay, Alibaba GMV data per company. Meituan take rate is estimate per media report.

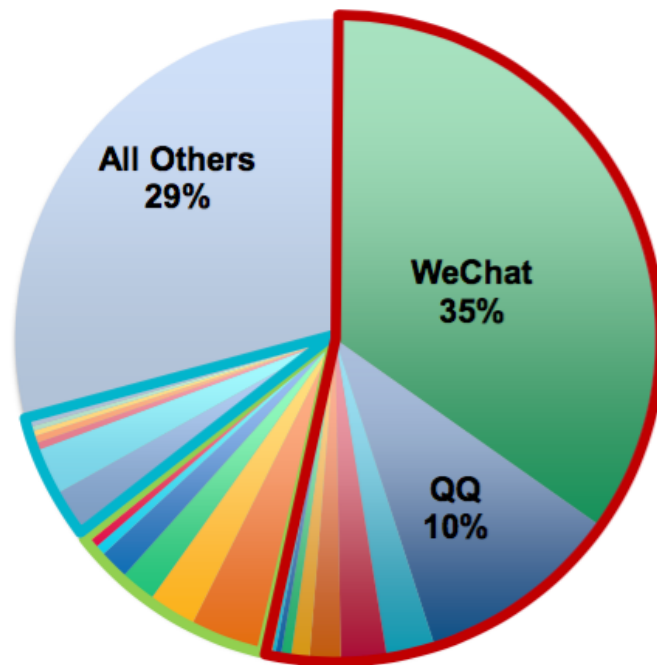
Hillhouse Capital

158

China Mobile Internet Usage Leaders...

Tencent + Alibaba + Baidu = 71% of Mobile Time Spent

Share of Mobile Time Spent, April 2016
Daily Mobile Time Spent = ~200 Minutes per User, Average



Tencent

- WeChat
- QQ
- QQ Browser
- Tencent Video
- Tencent News
- Tencent Games
- QQ Music
- JD.com
- QQ Reading

Alibaba

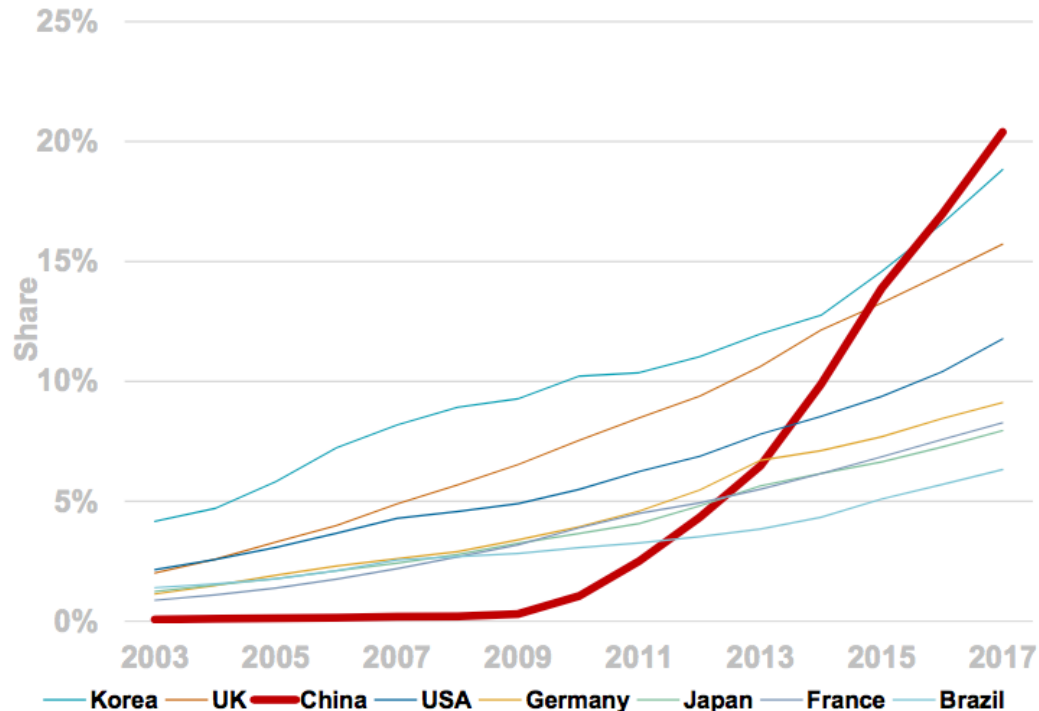
- UCWeb Browser
- Taobao
- Weibo
- YouKu Video
- Momo
- Shuqi Novel
- AliPay
- AutoNavi

Baidu

- Mobile Baidu
- iQiyi / PPS Video
- Baidu Browser
- Baidu Tieba
- 91 Desktop
- Baidu Maps
- All Other

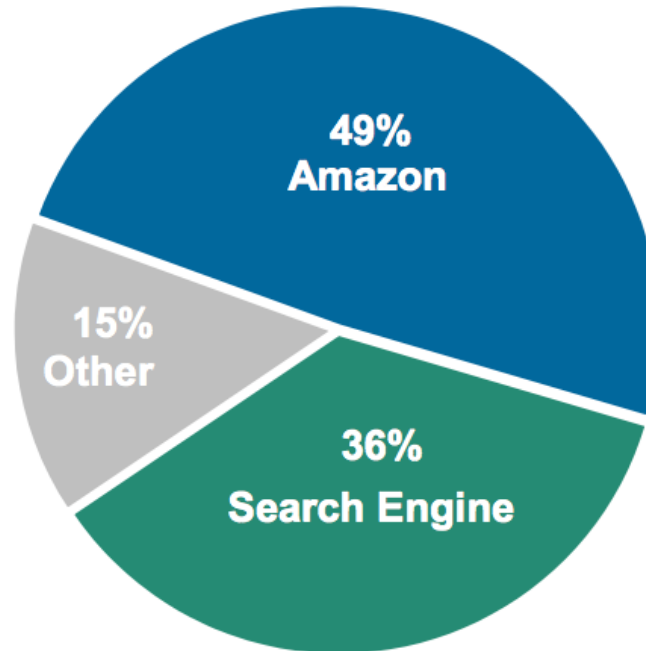
Worldwide E-Commerce Share Gains Continue... China @ 20% = Highest Penetration Rate + Fastest Growing

E-Commerce % of Retail Sales



Product Finding = Often Starts @ Search (Amazon + Google...)

Where Do You Begin Your Product Search?



Product Finding (Amazon) = Started @ Search...Fulfilled by Amazon

Product Search



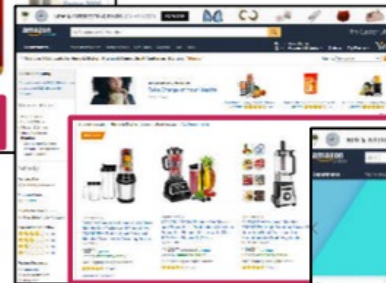
1-Click Purchasing



Prime Fulfillment



Sponsored Product Listings



Voice Search + Fulfillment



Product Finding (Google) = Started @ Search...Fulfilled by Others

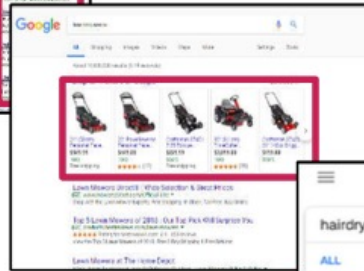
Organic Search



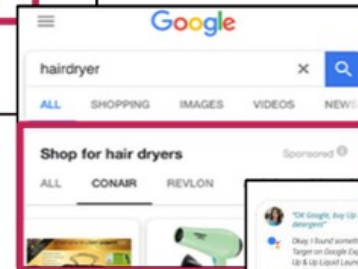
Paid Search



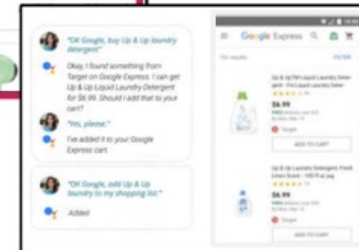
Google Shopping



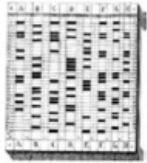
Product Listing Ads



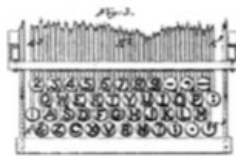
Shopping Actions



Human-Computer Interaction (1830s – 2015), USA = Touch 1.0 → Touch 2.0 → Touch 3.0 → Voice



Punch Cards for
Informatics
1832



QWERTY
Keyboard
1872



Electromechanical
Computer (Z3)
1941



Electronic Computer
(ENIAC)
1943



Paper Tape Reader
(Harvard Mark I)
1944



Mainframe Computers
(IBM SSEC)
1948



Trackball
1952



Joystick
1967



Microcomputers
(IBM Mark-8)
1974



Portable Computer
(IBM 5100)
1975



Commercial Use of
Window-Based GUI
(Xerox Star)
1981



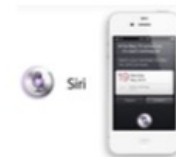
Commercial Use
of Mouse
(Apple Lisa)
1983



Commercial Use
of Mobile
Computing
(PalmPilot)
1996



Touch + Camera -
based Mobile
Computing
(iPhone 2G)
2007



Voice on Mobile
(Siri)
2011

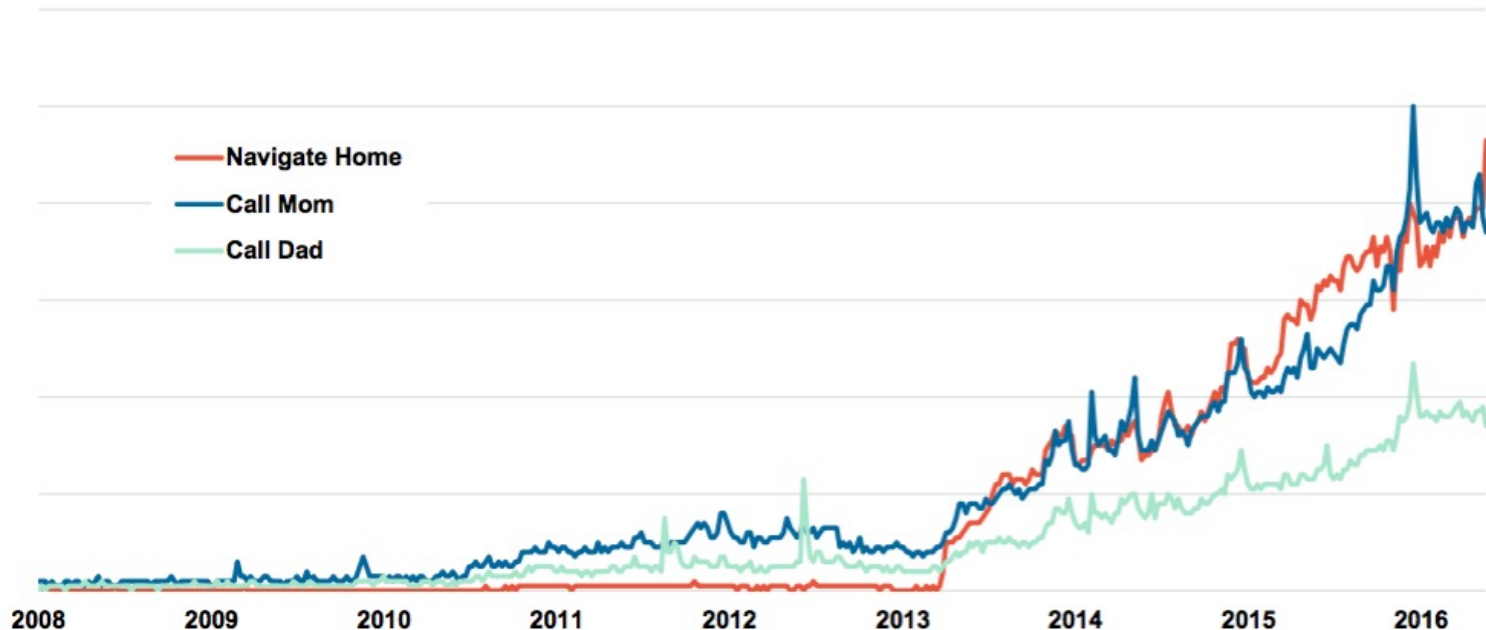


Voice on Connected /
Ambient Devices
(Amazon Echo)
2014

Google Voice Search Queries = Up >35x Since 2008 & >7x Since 2010, per Google Trends

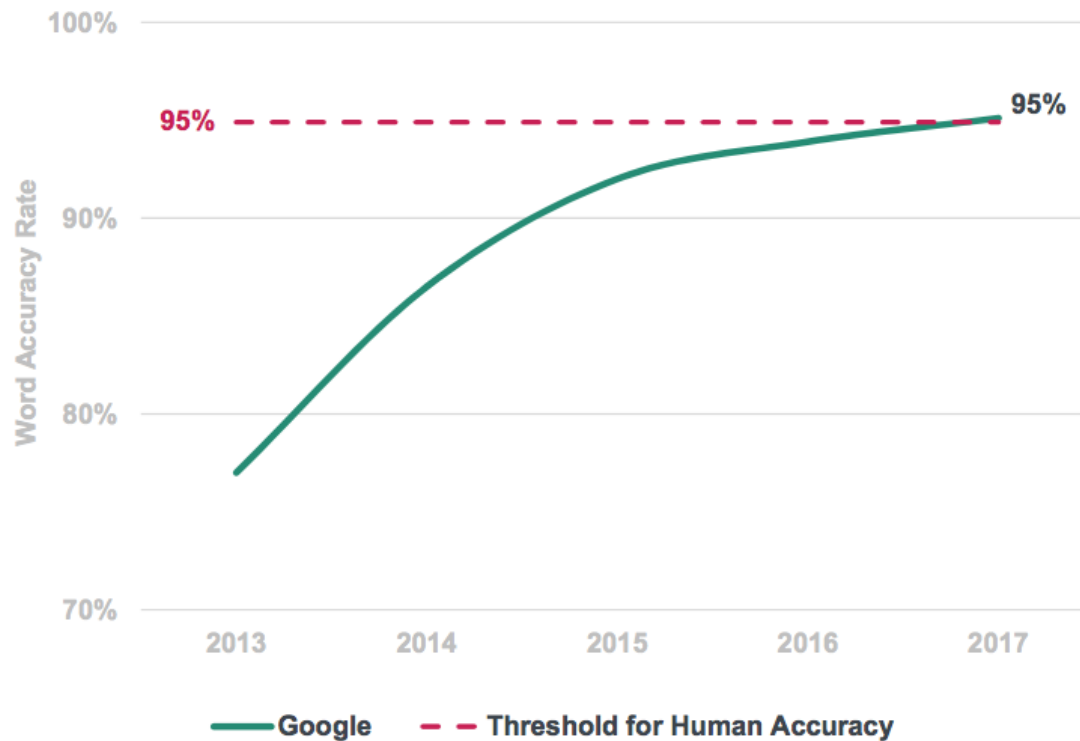
Google Trends imply queries associated with voice-related commands have risen >35x since 2008 after launch of iPhone & Google Voice Search

Google Trends, Worldwide, 2008 – 2016



Voice = Technology Lift Off...

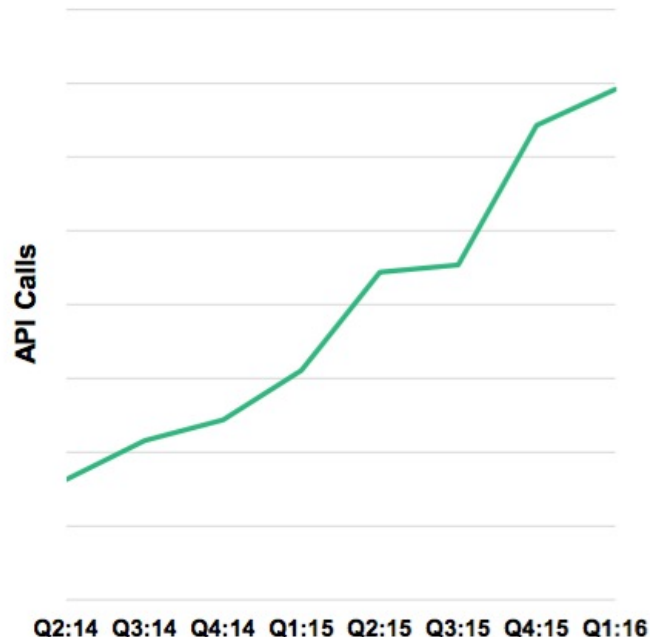
Google Machine Learning Word Accuracy



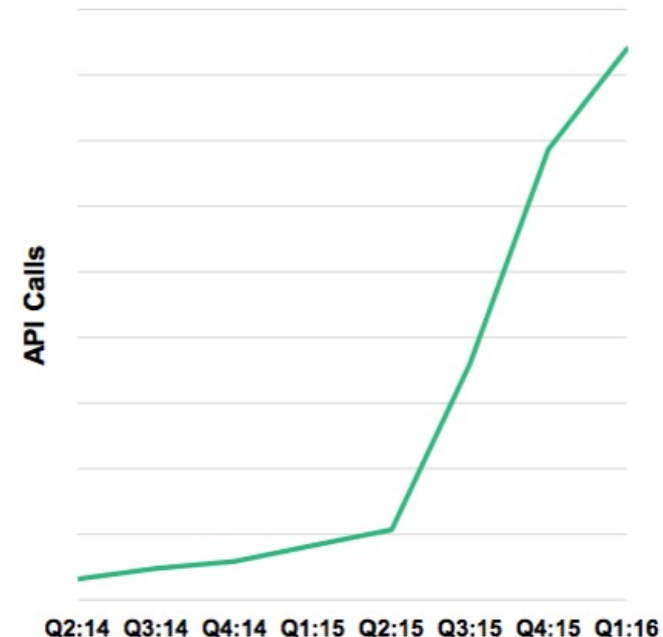
Baidu Voice = Input Growth >4x...Output >26x, Since Q2:14

Usage across all Baidu products growing rapidly...typing Chinese on small cellphone keyboard even more difficult than typing English...Text-to-Speech supplements speech recognition & key component of man-machine communications using voice

Baidu Speech Recognition Daily Usage by API Calls, Global, 2014 – 2016¹



Baidu Text to Speech (TTS) Daily Usage by API Calls, Global, 2014 – 2016²

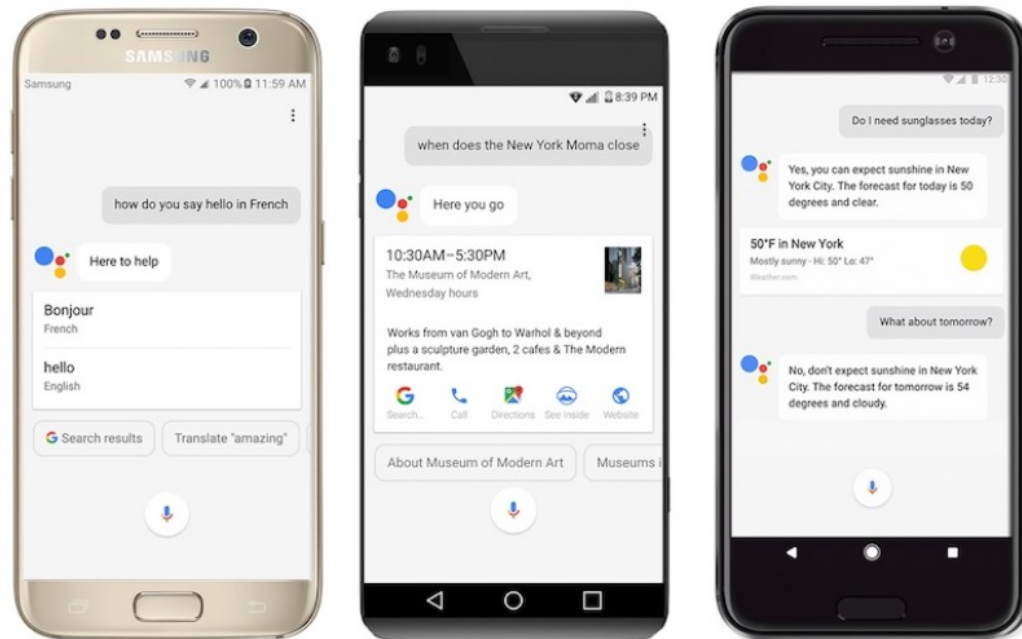


Voice-Based *Mobile* Platform *Front-Ends* = Voice Can Replace Typing

Google Assistant

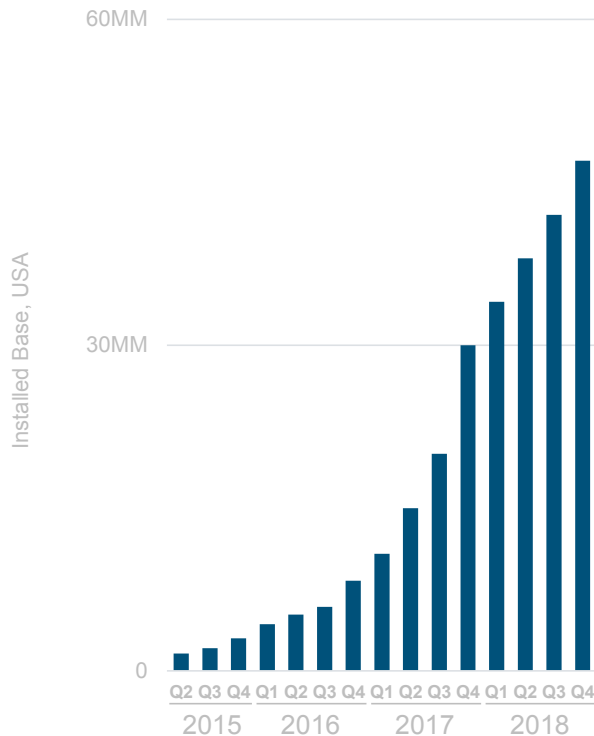
Nearly 70% of Requests are Natural / Conversational Language, 5/17

20% of Mobile Queries Made via Voice, 5/16

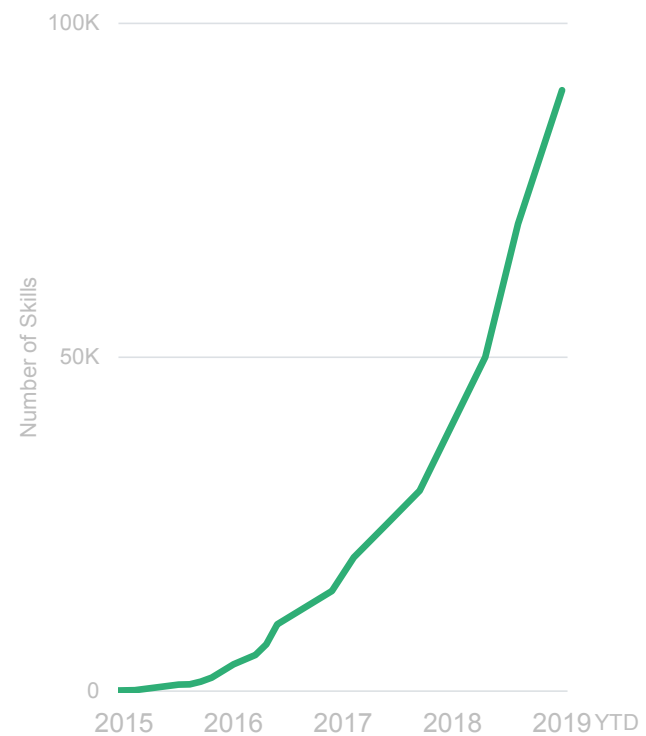


...Voice =
47MM Amazon Echo Base + ~2x in One Year

Amazon Echo Installed Base

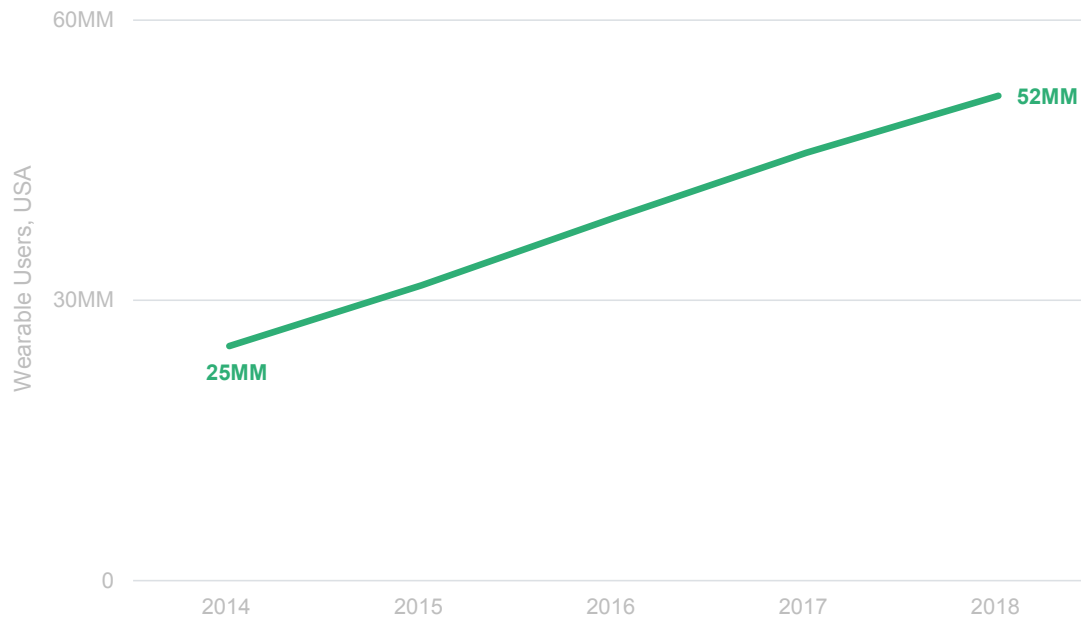


Amazon Echo Skills



Wearables =
52MM Users + ~2x in Four Years

Wearable Users, USA



Is it a Car...Is it a Computer?...

Is it a Phone...Is it a Camera?



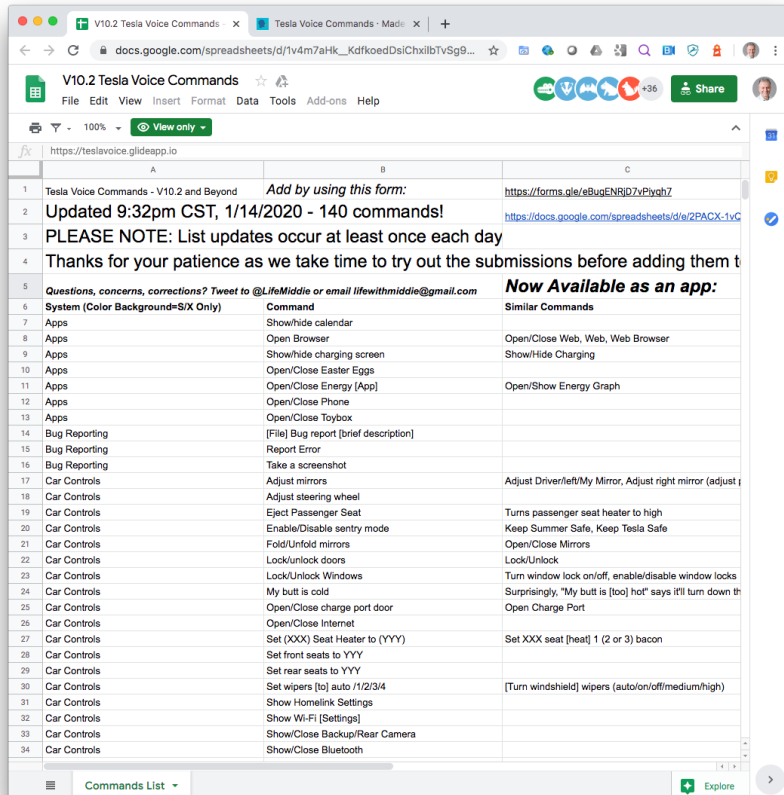
Is it a Car...Is it a Computer?



...One Can... Lock / Monitor / Summon One's Tesla from One's Wrist



Tesla Voice Commands



V10.2 Tesla Voice Commands

Updated 9:32pm CST, 1/14/2020 - 140 commands!

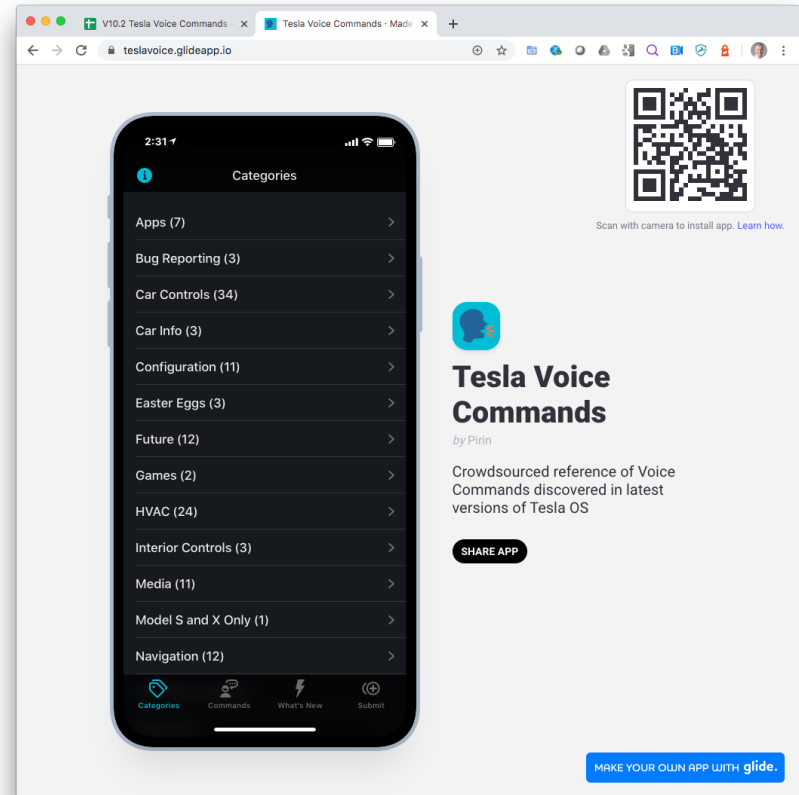
PLEASE NOTE: List updates occur at least once each day

Thanks for your patience as we take time to try out the submissions before adding them to the list

Questions, concerns, corrections? Tweet to @LifeMiddle or email lifewithmiddle@gmail.com

System (Color Background=S/X Only)	Command	Similar Commands
Apps	Show/hide calendar	
Apps	Open Browser	Open/Close Web, Web, Web Browser
Apps	Show/hide charging screen	Show/Hide Charging
Apps	Open/Close Easter Eggs	
Apps	Open/Close Energy [App]	Open/Show Energy Graph
Apps	Open/Close Phone	
Apps	Open/Close Toybox	
Bug Reporting	[File] Bug report [brief description]	
Bug Reporting	Report Error	
Bug Reporting	Take a screenshot	
Car Controls	Adjust mirrors	Adjust Driver/left/My Mirror, Adjust right mirror (adjust)
Car Controls	Adjust steering wheel	
Car Controls	Eject Passenger Seat	Turns passenger seat heater to high
Car Controls	Enable/Disable sentry mode	Keep Summer Safe, Keep Tesla Safe
Car Controls	Fold/Unfold mirrors	Open/Close Mirrors
Car Controls	Lock/unlock doors	Lock/Unlock
Car Controls	Lock/Unlock Windows	Turn window lock on/off, enable/disable window locks
Car Controls	My butt is cold	Surprisingly, "My butt is [too] hot" says it'll turn down the
Car Controls	Open/Close charge port door	Open Charge Port
Car Controls	Open/Close Internet	
Car Controls	Set (XXX) Seat Heater to (YYY)	Set XXX seat [heat] 1 (2 or 3) bacon
Car Controls	Set front seats to YYY	
Car Controls	Set rear seats to YYY	
Car Controls	Set wipers [to] auto /1/2/3/4	[Turn windshield] wipers (auto/on/off/medium/high)
Car Controls	Show Homelink Settings	
Car Controls	Show Wi-Fi [Settings]	
Car Controls	Show/Close Backup/Rear Camera	
Car Controls	Show/Close Bluetooth	

Commands List



<https://teslavoice.glideapp.io/>

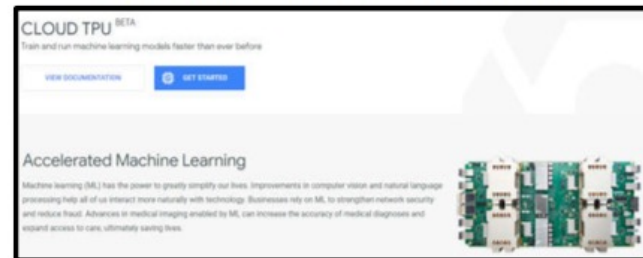
...Google = AI Platform Emerging from Google Cloud... Enabling Easier Data Processing / Collection for Others

Google Cloud AI Services / Infrastructure

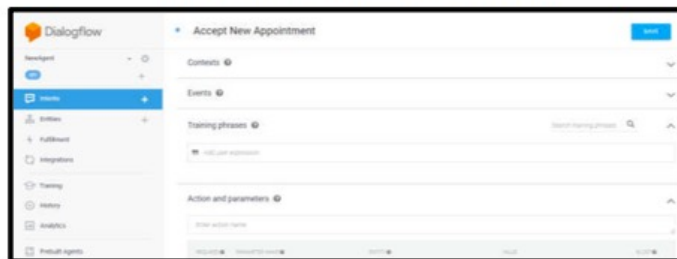
Google Cloud Vision API



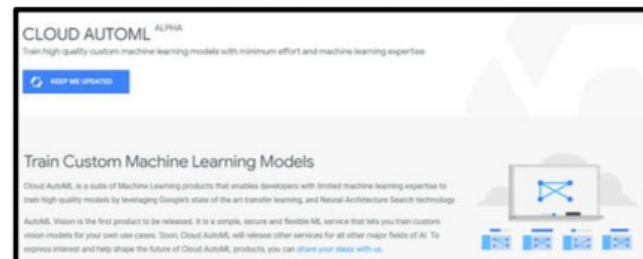
AI Hardware – Tensor Processing Units



Dialogflow Conversational Platform

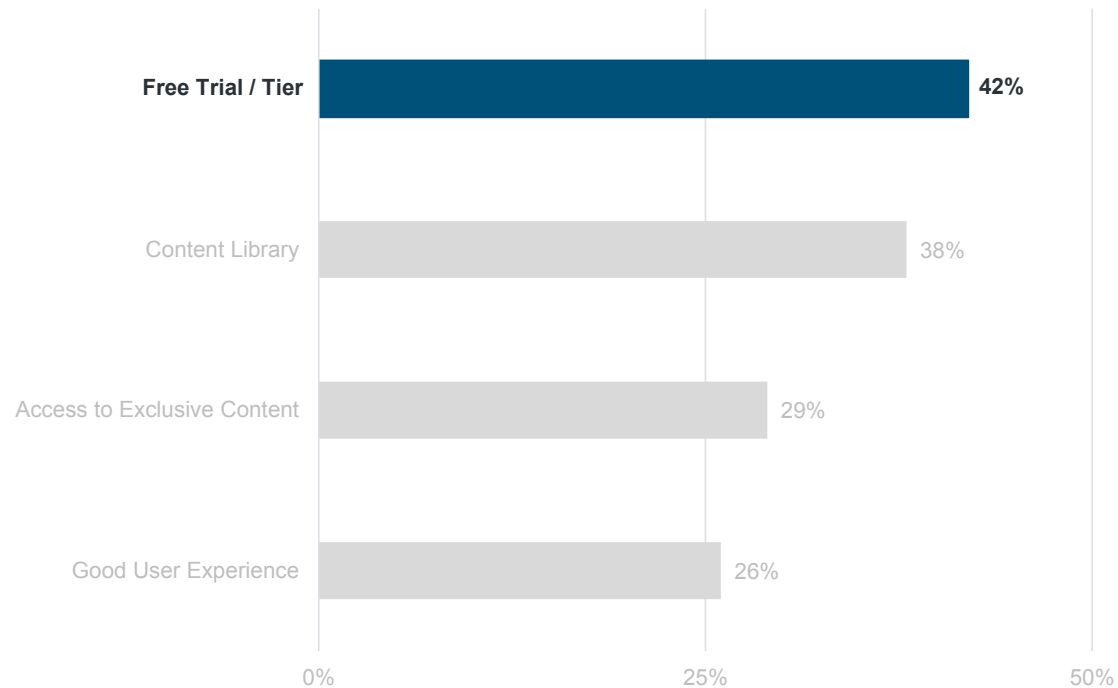


Cloud AutoML – Custom Models



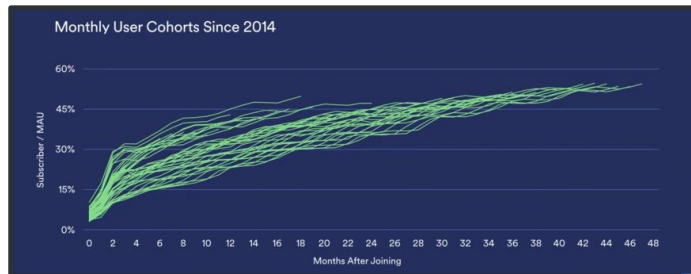
Effective + Efficient Marketing = Can Be Free Trial / Tier

Online Streaming – Reasons For Trying New Service



Happy Customers... Spotify = Free User Conversion to Paid Subscribers...

Free Ad-Supported Product...

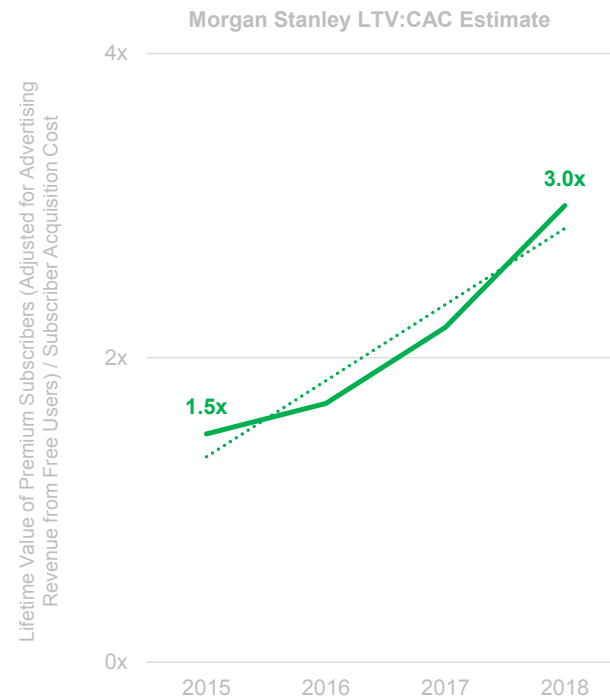


Our freemium model accounts for ~60% of our gross added premium subscribers... the ad-supported service is a subsidy program that offsets costs of new subscriber acquisition.

Developing a better user experience produces by far the most viral effect & impact when investing in growth. Engagement drives conversion from free consumption to paid subscription.

Barry McCarthy – CFO, Spotify, 3/18

...Rising LTV / Subscriber Acquisition Cost Ratio



...Happy Customers... Zoom = Free User Conversion to Paid Subscribers

Free to Join, When Paid User Hosts...



*..we really want to get customers to test our product...
It's really hard to get customers to
try Zoom without a freemium product...*

***We make our freemium product work so well...
If they like our product, very soon they are
going to pay for the subscription.***

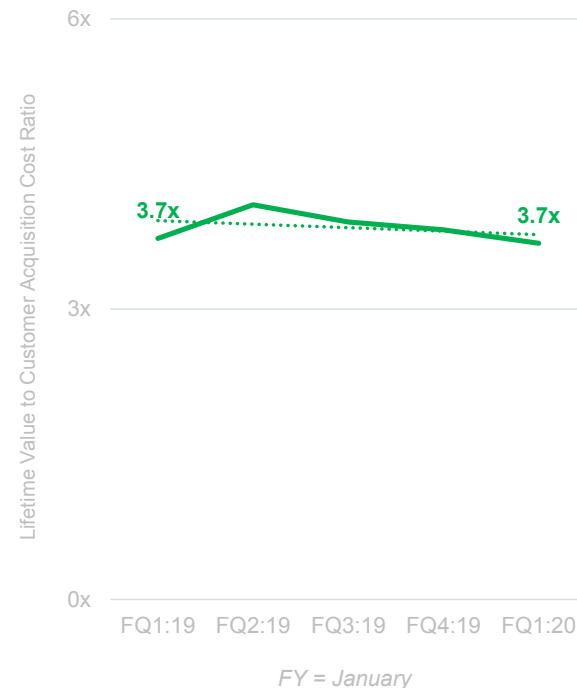
*The most important thing is to make sure the existing
customer [is] happy rather than chasing after new prospects.*

*Our NPS is in the 67-69 range vs. our peers in the 20's...
We do not want to spend money on [the]
marketing side to generate leads.*

Eric Yuan – Founder / CEO, Zoom, 8/17

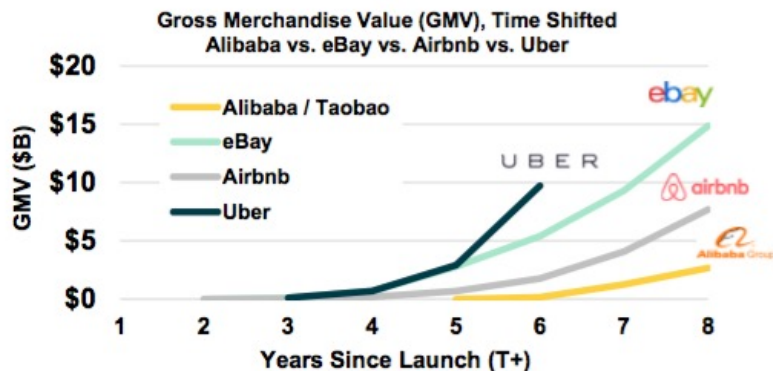
...High LTV / CAC Ratio*

Goldman Sachs Investment Research LTV:CAC Estimate



Current Generation of Internet Leaders = Growing Faster than Previous Generation

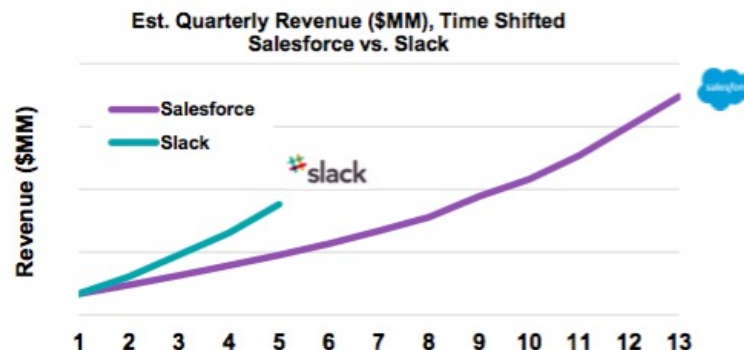
Marketplaces



Commerce



Enterprise



Global *Internet* Market Capitalization Leaders = USA Stable @ 18 of 30...China Stable @ 7 of 30

Rank 2019	Company	Region	Market Cap Value (\$B)		% Change
			6/7/19	6/7/16	
1	Microsoft	USA	\$1,007B	\$410B	+146%
2	Amazon	USA	888	343	+159%
3	Apple	USA	875	540	+62%
4	Alphabet	USA	741	497	+49%
5	Facebook	USA	495	340	+46%
6	Alibaba	China	402	195	+106%
7	Tencent	China	398	206	+93%
8	Netflix	USA	158	43	+266%
9	Adobe	USA	136	50	+174%
10	PayPal	USA	134	46	+190%
11	Salesforce	USA	125	56	+123%
12	Booking.com	USA	77	67	+15%
13	Uber	USA	75	--	--
14	Recruit Holdings	Japan	52	20	+167%
15	ServiceNow	USA	51	12	+316%
16	Workday	USA	48	16	+197%
17	Meituan Dianping	China	44	--	--
18	JD.com	China	39	32	+22%
19	Baidu	China	38	60	(36%)
20	Activision Blizzard	USA	35	28	+25%
21	Shopify	Canada	34	2	+1,297%
22	NetEase	China	33	23	+44%
23	eBay	USA	33	28	+19%
24	Atlassian	Australia	32	5	+509%
25	MercadoLibre	Argentina	30	6	+388%
26	Twitter	USA	29	11	+173%
27	Square	USA	29	3	+808%
28	Electronic Arts	USA	29	23	+25%
29	Xiaomi	China	28	--	--
30	Spotify	Sweden	25	--	--
Total			\$6,119	\$3,064	

Global Market Capitalization Leaders = USA Stable @ 23 of 30...Technology Stable @ 9 of 30

Rank 2019	Company	Sector	Region	Market Cap Value (\$B)		% Change
				6/7/19	6/7/16	
1	Microsoft	Technology	USA	\$1,007B	\$410B	+146%
2	Amazon	Technology	USA	888	343	+159%
3	Apple	Technology	USA	875	540	+62%
4	Alphabet	Technology	USA	741	497	+49%
5	Berkshire Hathaway	Financial Services	USA	505	350	+44%
6	Facebook	Technology	USA	495	340	+46%
7	Alibaba	Technology	China	402	195	+106%
8	Tencent	Technology	China	398	206	+93%
9	Visa	Financial Services	USA	372	192	+94%
10	Johnson & Johnson	Healthcare	USA	368	318	+16%
11	JPMorgan	Financial Services	USA	354	239	+48%
12	Exxon Mobil	Energy	USA	316	371	(15%)
13	Nestlé	Food / Beverages	Switzerland	306	230	+33%
14	Walmart	Retail	USA	303	221	+37%
15	ICBC	Financial Services	China	285	224	+27%
16	Procter & Gamble	Home Goods	USA	273	220	+24%
17	Mastercard	Financial Services	USA	271	106	+156%
18	Bank of America	Financial Services	USA	262	149	+76%
19	Royal Dutch Shell	Energy	Netherlands	259	198	+31%
20	Samsung	Technology	South Korea	249	166	+50%
21	Disney	Media	USA	248	160	+55%
22	Cisco	Technology	USA	239	146	+64%
23	Pfizer	Pharmaceuticals	USA	238	212	+12%
24	AT&T	Telecom	USA	237	242	(2%)
25	Verizon	Telecom	USA	237	207	+15%
26	UnitedHealth	Healthcare	USA	235	131	+79%
27	Roche	Healthcare	Switzerland	233	224	+4%
28	Chevron	Energy	USA	231	191	+21%
29	Coca-Cola	Food / Beverages	USA	220	196	+12%
30	Home Depot	Retail	USA	217	161	+35%
Total				\$11,264	\$7,385	

USA = 60% of Most Highly Valued *Tech* Companies Founded By... 1st or 2nd Generation Americans...1.9MM Employees, 2018

Immigrant Founders / Co-Founders of Top 25 USA Valued Public *Tech* Companies, Ranked by Market Capitalization

Rank	Company	Mkt Cap (\$B)	LTM Rev (\$B)	Employees (K)	Founder / Co-Founder (1st or 2nd Gen Immigrant)	Generation
1	Microsoft	\$1,007B	\$122B	131K	--	--
2	Amazon	888	242	648	Jeff Bezos	2nd, Cuba
3	Apple	875	258	132	Steve Jobs	2nd, Syria
4	Alphabet / Google	741	142	99	Sergey Brin	1st, Russia
5	Facebook	495	59	36	Eduardo Saverin	1st, Brazil
6	Cisco	239	51	74	--	--
7	Intel	206	71	107	--*	--
8	Oracle	182	40	137	Larry Ellison / Bob Miner	2nd, Russia / 2nd, Iran
9	Netflix	158	17	7	--	--
10	Adobe	136	10	21	--	--
11	PayPal	134	16	22	Max Levchin / Luke Nosek / Peter Thiel / Elon Musk***	1st, Ukraine / 1st, Poland / 1st, Germany / 1st, South Africa
12	Salesforce	125	14	35	--	--
13	IBM	118	79	351	Herman Hollerith	2nd, Germany
14	Texas Instruments	104	16	30	Cecil Green / J. Erik Jonsson	1st, UK / 2nd, Sweden
15	NVIDIA	89	11	13	Jensen Huang	1st, Taiwan
16	Qualcomm	84	21	35	Andrew Viterbi	1st, Italy
17	Booking.com	77	14	25	--	--
18	Uber	75	12	22	Garrett Camp	1st, Canada
19	Automatic Data Processing	73	14	57	Henry Taub	2nd, Poland
20	VMware	69	9	23	Edouard Bugnion	1st, Switzerland
21	Intuit	67	7	9	--	--
22	ServiceNow	51	3	8	--	--
23	Workday	48	3	11	Aneel Bhusri	2nd, India
24	Micron	38	30	36	--	--
25	Cognizant	36	16	282	Francisco D'Souza / Kumar Mahadeva	1st, India** / 1st, Sri Lanka

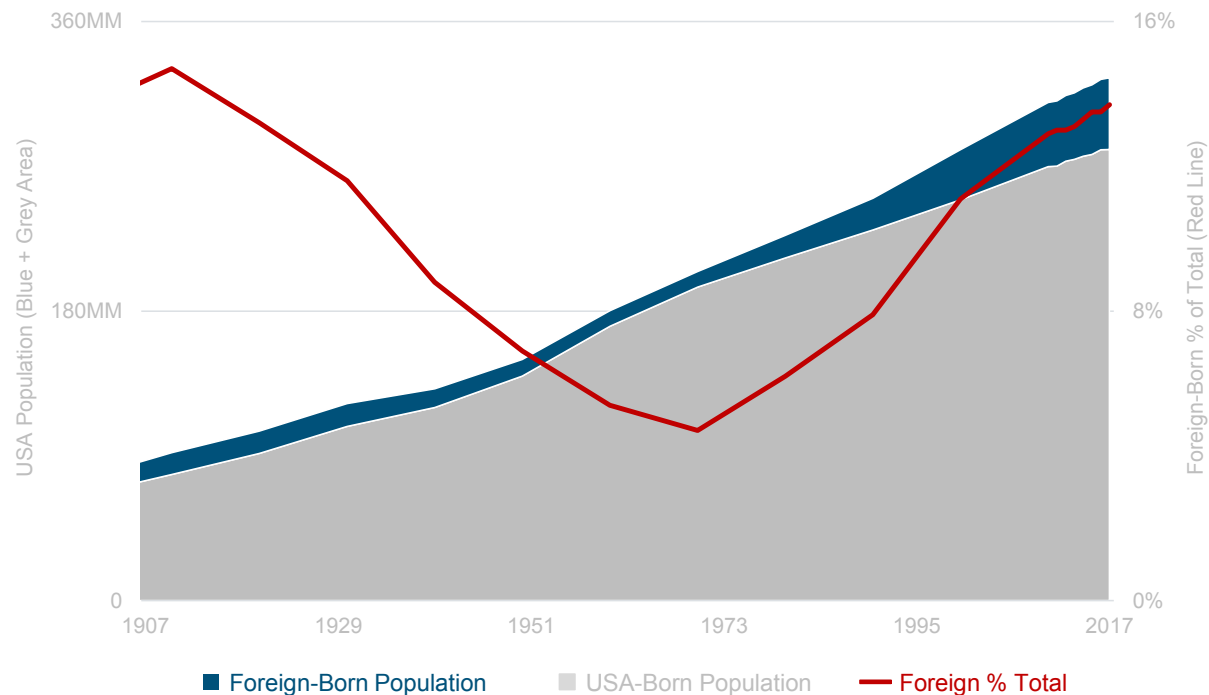
USA = Many Highly Valued Private Tech Companies Founded By... 1st Generation Immigrants

Company	Founder / Co-Founder	Country of Origin	Valuation (\$B)
WeWork	Adam Neumann	Israel	\$47B
SpaceX	Elon Musk	South Africa	31
Stripe	John Collison	Ireland	23
	Patrick Collison		
Palantir	Peter Thiel	Germany	21
Epic Games	Mark Rein	Canada	15
DoorDash	Tony Xu	China	13
Wish	Peter Szulczewski	Canada	9
	Danny Zhang		
Instacart	Apoorva Mehta	India	8
Slack	Stewart Butterfield	Canada	7
	Serguei Mourachov	Russia	
UIPath*	Cal Henderson	UK	7
	Daniel Dines	Romania	
Tanium	Marius Tirca		7
	David Hindawi	Iraq	
Unity Technologies	David Helgason	Iceland	6
	Nicholas Francis	Denmark	
Robinhood	Joachim Ante	Germany	6
	Baiju Bhatt	India	
Compass	Vlad Tenev	Bulgaria	4
	Ori Allon	Israel	
Credit Karma	Kenneth Lin	China	4
Houzz	Adi Tatarko	Israel	4
	Alon Cohen		
Snowflake	Marcin Zukowski	Netherlands	4
	Benoit Dageville	France	
	Thierry Cruanes	France	
	Bipul Sinha	India	
Rubrik	Arvind Nithrakashyap	India	3
	Arvind Jain	India	
	Soham Mazumdar	India	
	Tim Kentley-Klay	Australia	
Oscar Health	Mario Schlosser	Germany	3

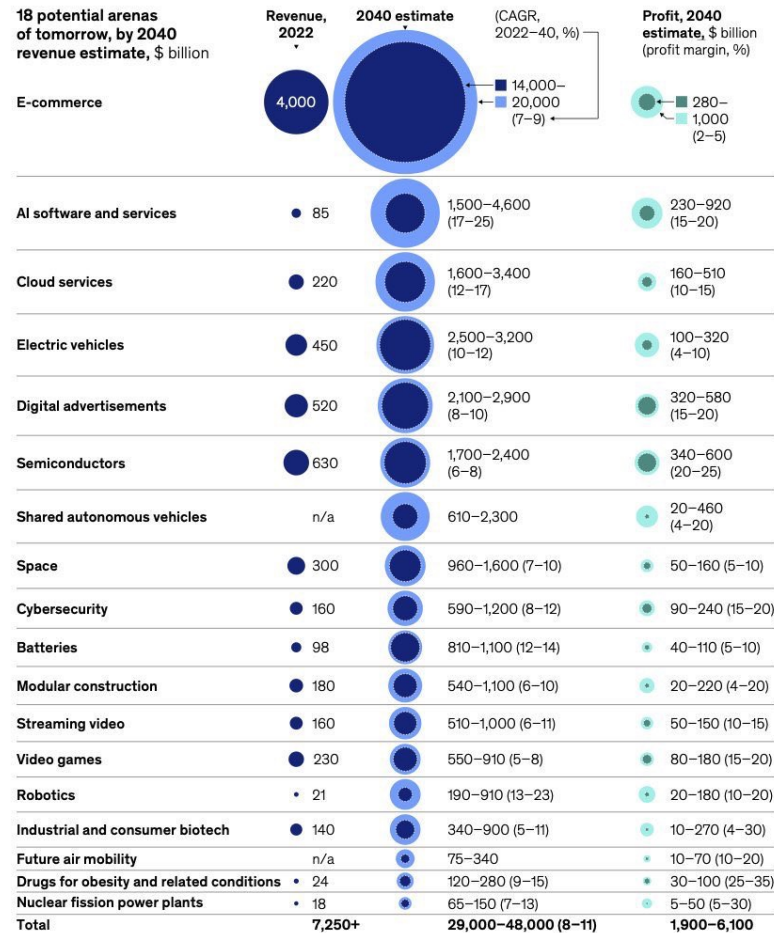
Company	Founder / Co-Founder	Country of Origin	Valuation (\$B)
Crowdstrike	Dmitri Alperovitch	Russia	\$3B
Affirm	Max Levchin	Ukraine	3
Databricks	Ali Ghodsi	Sweden	3
	Matei Zaharia	Romania	
Nuro	Ion Stoica	Romania	3
	Jiajun Zhu	China	
	Dave Ferguson	New Zealand	
	Ankur Kothari	India	
Automation Anywhere	Mihir Shukla	India	3
	Neeti Metha Shukla	India	
Confluent	Rushabh Parm	India	3
	Jun Rao	China	
Roblox	Neha Narkhede	India	2
	David Baszucki	Canada	
Medallia	Borge Hald	Norway	2
Lime	Toby Sun	China	2
Zume Pizza	Brad Bao		2
Gusto	Alex Garden	Canada	2
Lemonade	Tomer London	Israel	2
	Shai Wininger	Israel	
LegalZoom	Daniel Schreiber		2
	Brian Lee	South Korea	
Avant	AI Goldstein	Uzbekistan	2
	John Sun	China	
Apttus	Paul Zhang	China	2
	Krik Krappe	UK	
Postmates	Bastian Lehmann	Germany	2
Sprinklr	Ragy Thomas	India	2
Cloudflare	Michelle Zatlyn	Canada	2
Carta	Manu Kumar	India	2
ZocDoc	Oliver Kharraz	Germany	2
Warby Parker	Dave Gilboa	Sweden	2
Carbon3D	Alex Ermoshkin	Russia	2
Pony.ai	James Peng	China	2
	Tiancheng Lo		
ServiceTitan	Ara Mahdessian	Iran	2
	Vahe Kuzoyan	Armenia	
Segment	Ilya Volodarsky	Russia	2
Quanergy	Tianyue Yu	China	2

USA = 14% of Population Foreign-Born & Rising... Near All-Time High (1910) @ 15%

USA Population vs. Foreign-Born % of Total Population



The 18 potential arenas of tomorrow could generate \$29 trillion to \$48 trillion in revenues and \$2 trillion to \$6 trillion in profits.

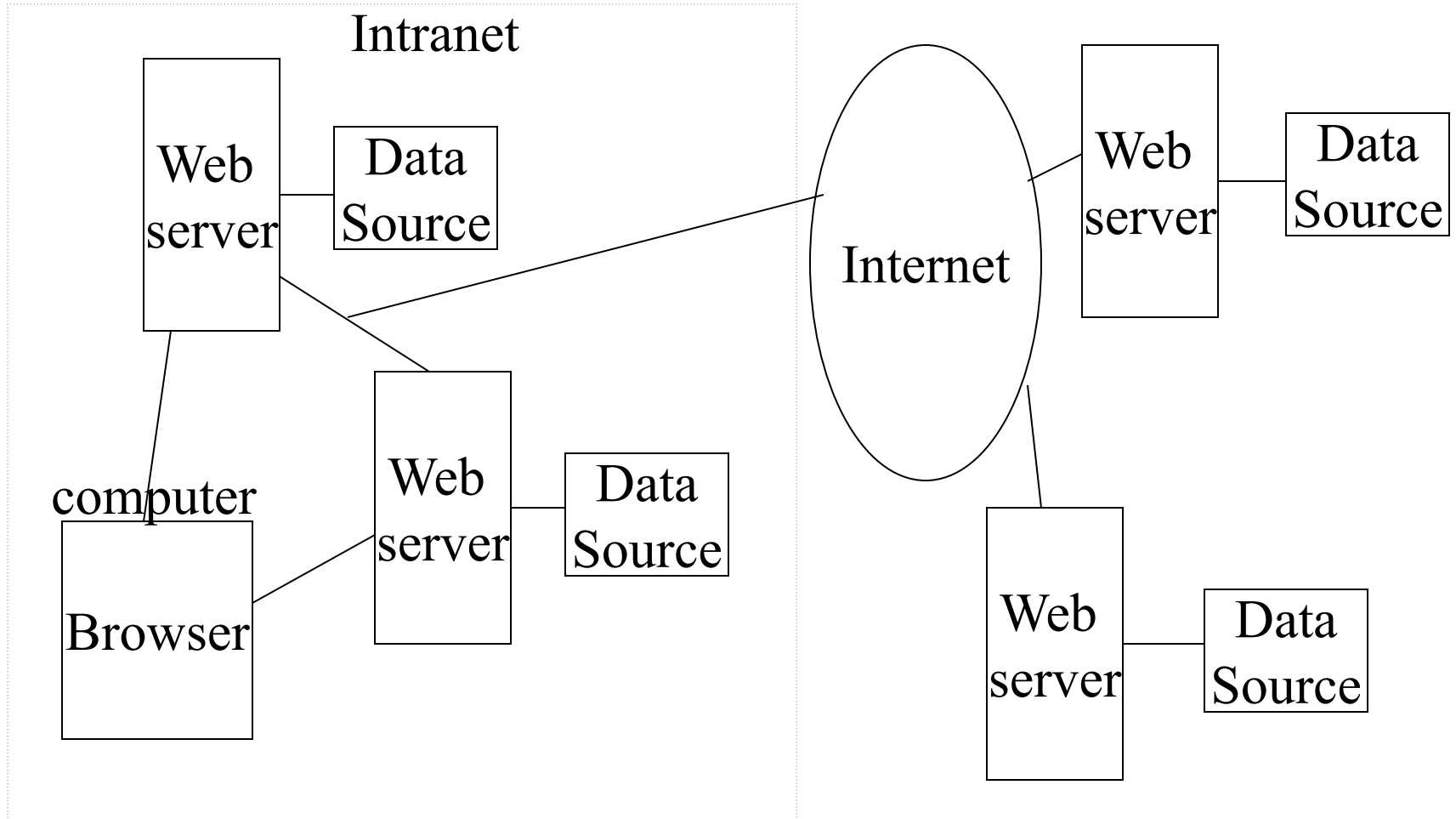


Source: The next big arenas of competition, McKinsey Global Institute, October 2024

The World Wide Web (WWW)

- Material relevant to exams starts HERE with this slide.

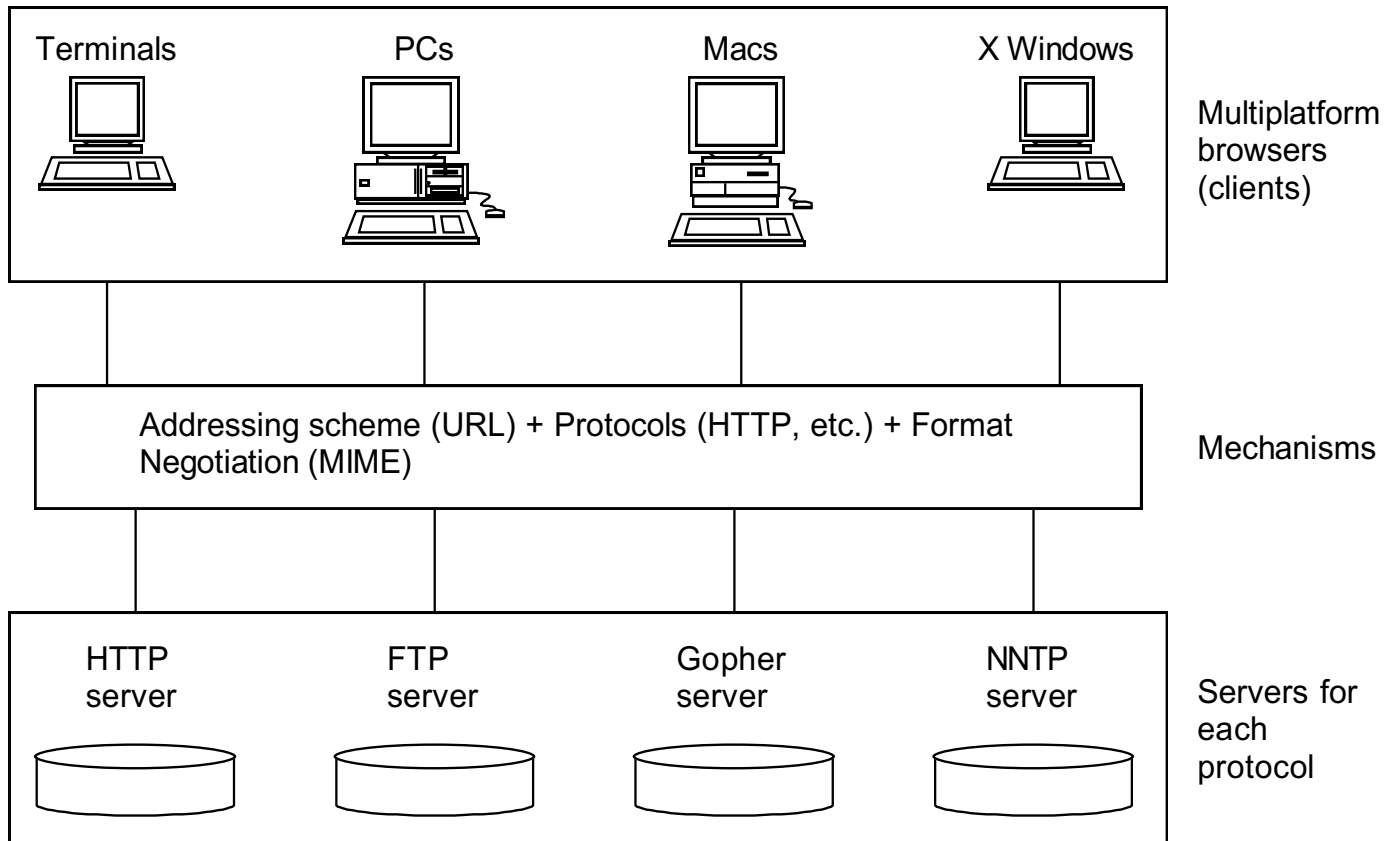
Graphical View of the WWW



Major Technology Components

- **Client/server architecture**
 - where client programs interact with web servers
- **Network protocol**
 - HTTP, Hypertext Transfer Protocol, is the language understood by browsers and web servers
 - designed to move quickly from document to document
- **Addressing system** (Uniform Resource Locators)
 - `http://domain/directory/file.html`
- **Markup Language**
 - every web server understands and every browser displays
 - includes support for HyperText and multimedia

Client/Server Architecture Model



The WWW Server

- Web browsers and Web servers communicate according to a protocol known as HTTP (HyperText Transfer Protocol)
 - The current HTTP protocol is version 2.0
- The Web server is a software system running on a machine often called the Web server, don't confuse them
- A web server can
 - receive and reply to HTTP requests
 - retrieve documents from specified directories
 - run programs in specified directories
 - handle limited forms of security
- A web server does not
 - know about the contents of a document, links in a document, images in a document or whether a particular file, e.g. a *.gif file, is in the correct format

Uniform Resource Locator (URL)

- A mechanism whereby an Internet resource can be specified in a single line of ASCII text
- See **RFC 1738**: <http://www.faqs.org/rfcs/rfc1738.html>

URL

Refers to:

`file://pub/xt.ps`

a PostScript file in directory
pub on your local machine

`ftp://usc.edu/docs/sweng.txt`

file `sweng.txt` in directory `docs`
on `usc.edu`, an anonymous ftp site

`http://nunki.usc.edu/mydocs/book.doc`

a file in directory `mydocs` on
machine `nunki.usc.edu`, a WWW site

`news:comp.compilers`

the newsgroup `computers.compilers`

`mailto:horowitz@usc.edu` an e-mail address

General Description of a URL

1. **Scheme** followed by a colon http:,ftp:, news:,mailto:,wais:,telnet:
2. **Double slash** (optional. Required for http, ftp) //
3. Internet **domain** name e.g., www.usc.edu
4. **Port** number (optional; e.g., www.usc.edu:8081)
Standard or default port numbers:

---	ftp is 21	gopher is 70
---	telnet is 23	http is 80
---	smtp is 25	nnntp is 119
---	imap is 143	secure nnntp is 563
---	pop3 is 110	secure pop3 is 995
5. **Path** e.g., /pub/docs

URL Character Set

- RFC 1738, Dec. 1994 defines the URL character set as "...Only alphanumerics [0-9a-zA-Z], the special characters "\$-_.+!*'(),", **[not including the quotes]**, and reserved characters used for their reserved purposes may be used unencoded within a URL."
- However, HTML supports ISO-8859-1 (ISO-Latin) character set
 - HTML 4.x extends the character set to all of Unicode
- Therefore, in URLs an escape mechanism is used, % followed by two hex digits
- Characters that should be encoded include:
%, /, ., .., #, ?, ;, :, \$, +, @, &, =
- Here are some encoded values for so-called "unsafe" characters

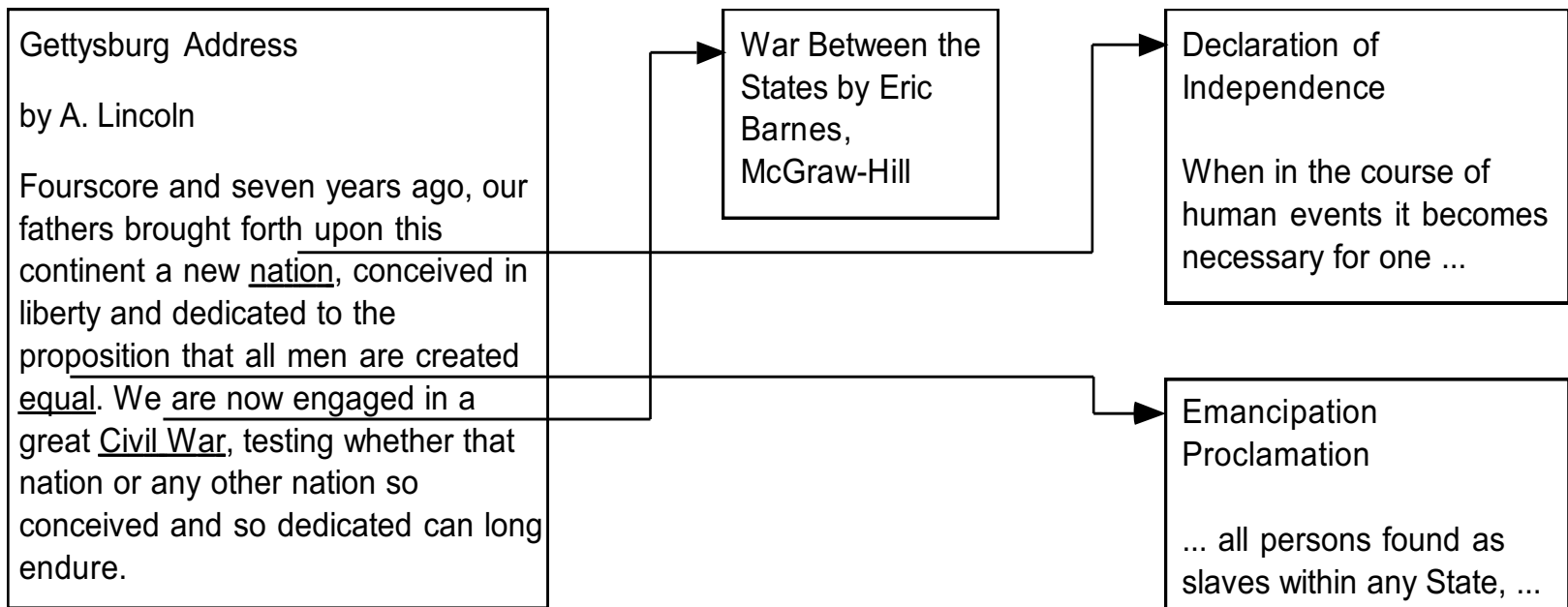
~	%7E		%7C
SPACE	%20	\	%5C
%	%25	^	%5E
&	%26	[	%5B
=	%3D	]	%5D
?	%3F	#	%23
{	%7B	>	%3E
}	%7D	<	%3C

Markup Languages

- HTML - hypertext markup language, specifies document layout and the specification of hypertext links to text, graphics and other types of objects
- Browsers display text and graphics using the markup as guidance
- However, HTML is *not* like a word processing program, e.g., Microsoft Word or WordPerfect, and *not* like a page description languages, e.g., postscript
 - as a result, translation into HTML can produce a result that does not look exactly like the original

What is HyperText?

- Regular text, with the additional feature of links to related documents
- As you read documents and follow links, you traverse a “web” of interconnections

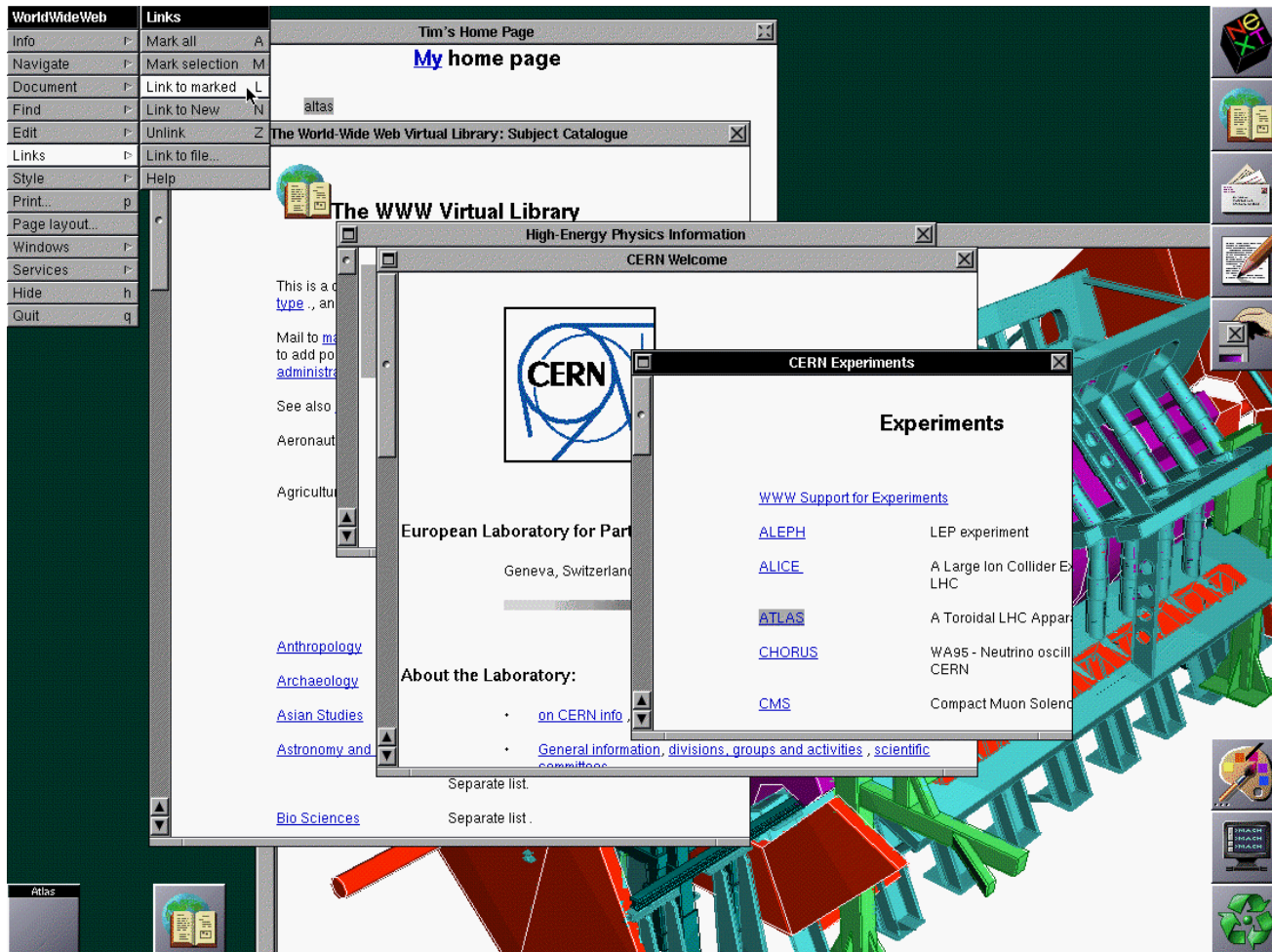


Early History of the WWW

- 1989-1990 Tim Berners-Lee conceives the WWW at CERN in Geneva
- 1990 Berners-Lee releases WWW prototype on NeXt computer
- 1992 Release of source code for line mode browser, lynx and HTTP
- 1993 Mosaic browser from NCSA is released
- 1993 WWW internet traffic now measures 1% of NSF backbone
- 12/94 Netscape Navigator 1.0 is released
World Wide Web Consortium formed
- 1995 Microsoft Windows 95 and Internet Explorer 1.0 released
- 1995 Java is released
- 1998 Google is started
- 1999-2001 A burst of Internet start-up companies which flamed out because they were not profitable. Also known as the "Internet Bubble."
- 2004 Firefox 1.0 is released
- 2005 YouTube is founded
- 2008 Google Chrome 1.0 is released

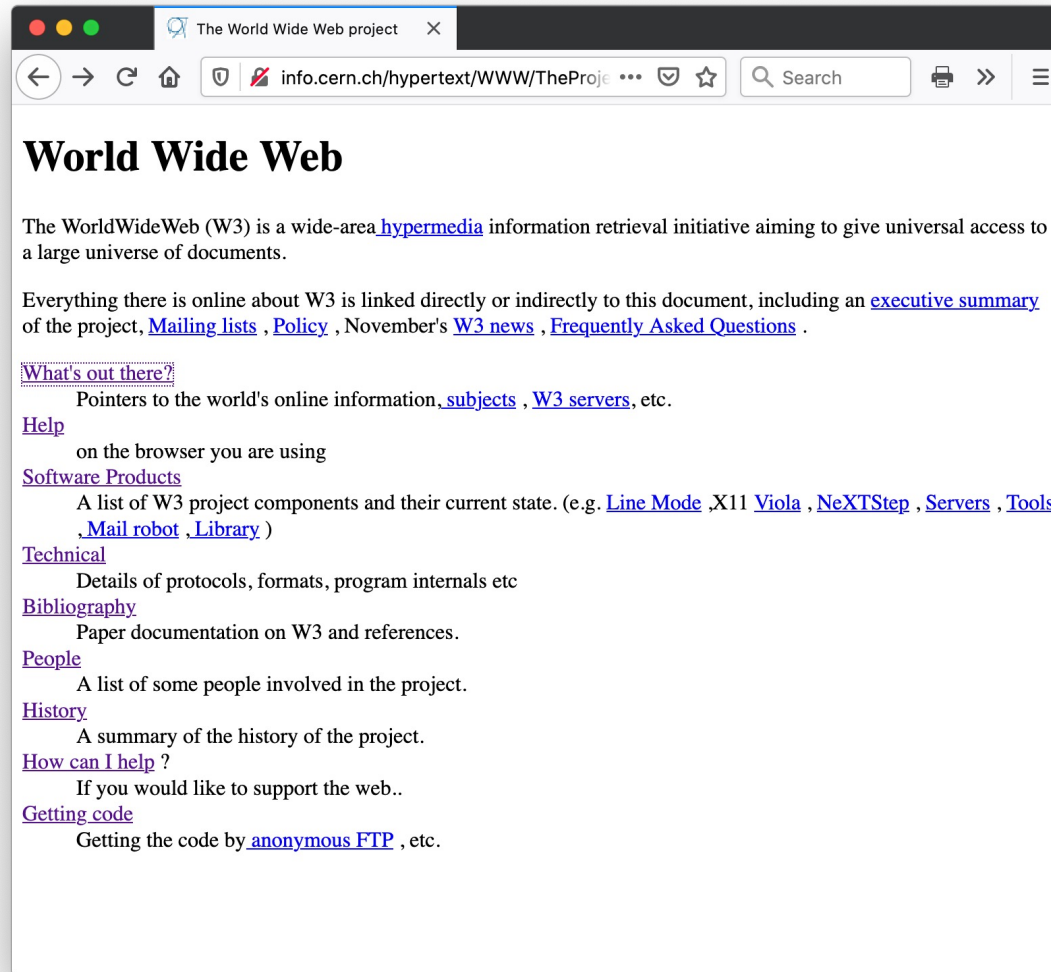
First Web Communication (Dec 1990)

See <http://www.w3.org/History.html> and tim Berners-Lee's presentation at the 10th anniversary, <http://www.w3.org/2004/Talks/w3c10-HowItAllStarted/?n=1>



Original WWW "The Project" site at CERN

`http://info.cern.ch/hypertext/WWW/TheProject.html`



London Olympics (July 2012)

See <http://www.zdnet.com/article/web-inventor-tim-berners-lee-stars-in-olympics-opening-ceremony/>

<https://www.youtube.com/watch?v=KW6ivwDcOY4>



Sir Tim Berners-Lee live-tweets during the 2012 Olympics opening ceremony, with a NeXT Cube by his side

WWW Consortium

- Founded in 1994, headed by Tim Berners-Lee, <http://www.w3.org>
- Goal: “to lead the World Wide Web to its full potential by developing common protocols that promote its evolution and ensure its interoperability.”
- Many of the technologies guided by the WWW consortium will be discussed this semester:
 - HTML, Style Sheets, Document Object Model, international character sets, HTTP, XML, etc.